The Role of Scaling Behavior in Time-Series Analysis and Predictability

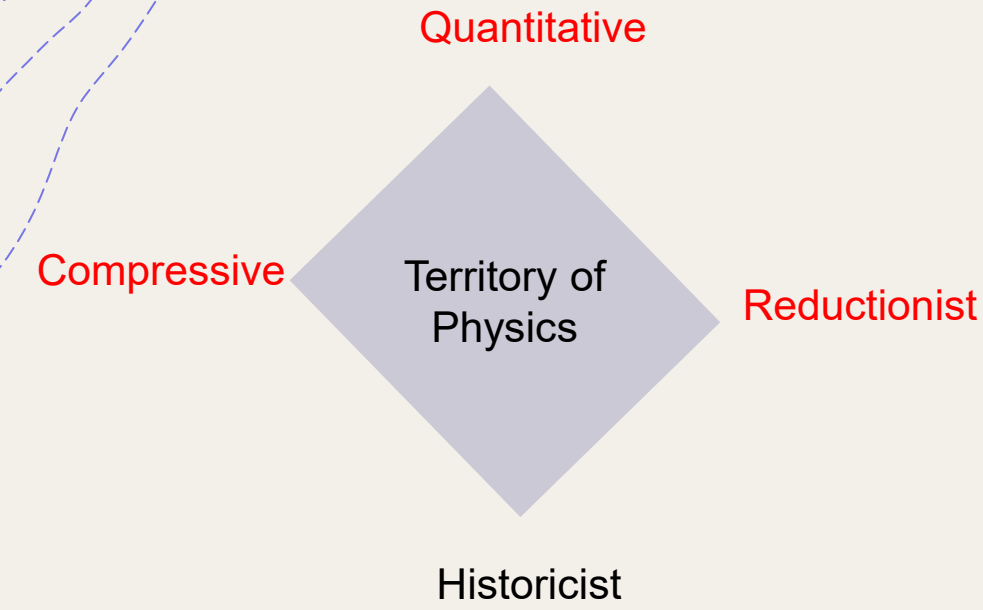
- Ali Khabazian
- Supervisor: Dr. Ali Hosseiny

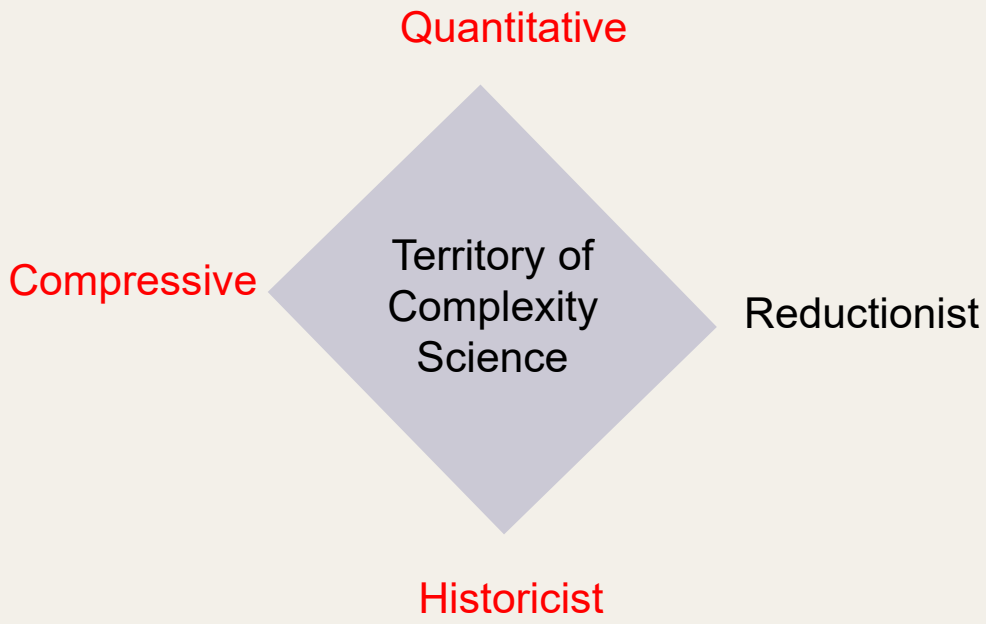
- August - 2026

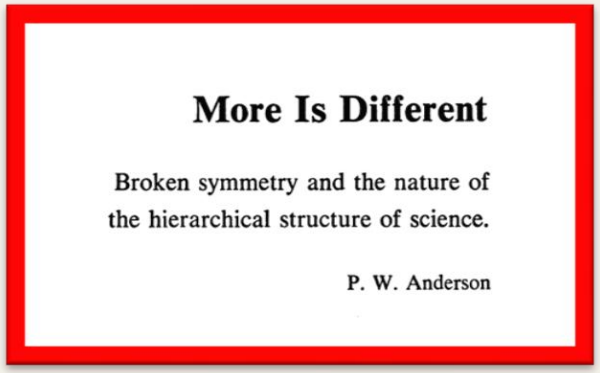
- **Contents:**

I. What is Complexity
II. Stylized Facts in Financial Markets
III. Scaling Laws
IV. Linear Models in Financial Market Analysis
V. Deep Learning Models in Financial Market Analysis
VI. Simulations and Results
VII. Hints for Further Studies

- The Complexity Science and the Disciplines







More Is Different

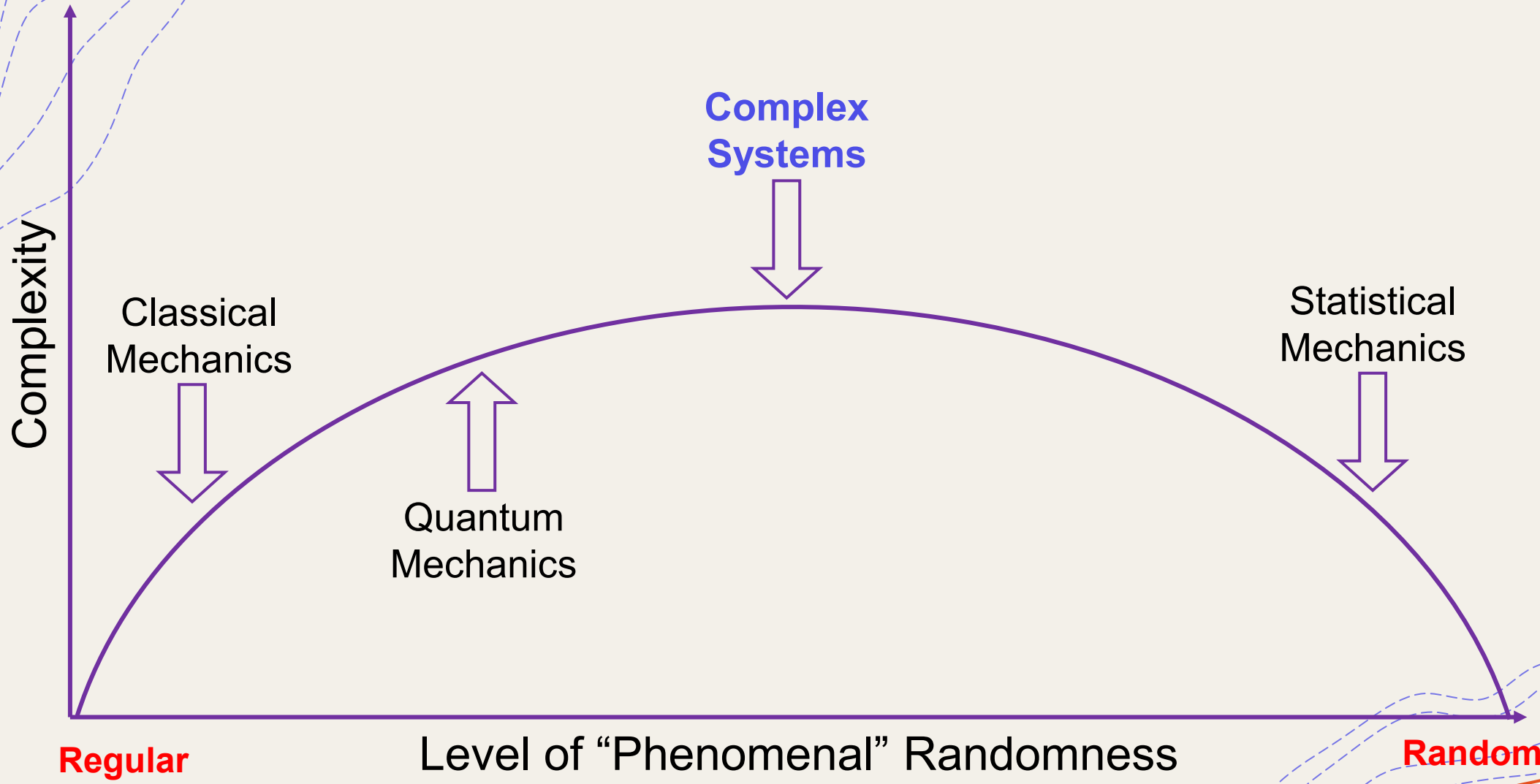
Broken symmetry and the nature of the hierarchical structure of science.

P. W. Anderson

Fig 1 – The Foundation paper in the field of complex systems

- Complex Systems Encode Adaptive Histories.

- The Complex Domain



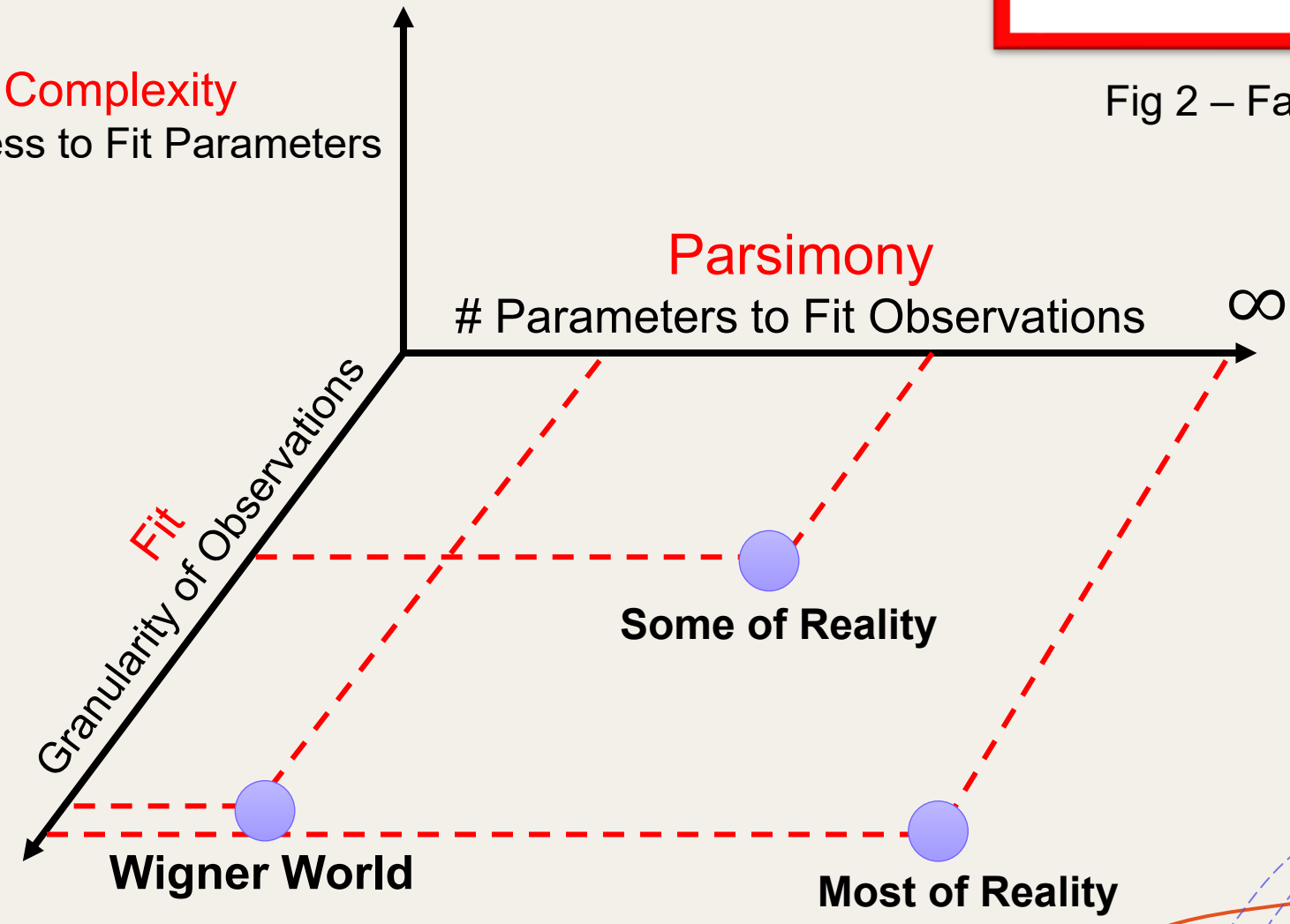
- Complexity Epistemology

The Unreasonable Effectiveness of Mathematics in the Natural Sciences

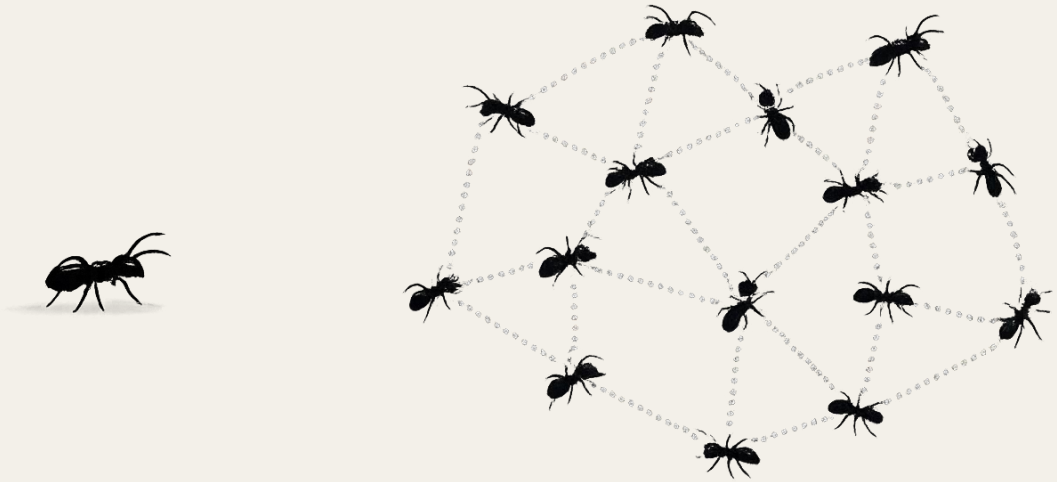
Richard Courant Lecture in Mathematical Sciences delivered at New York University, May 11, 1959

EUGENE P. WIGNER
Princeton University

Fig 2 – Famous paper by Wigner



- Complexity & Emergence

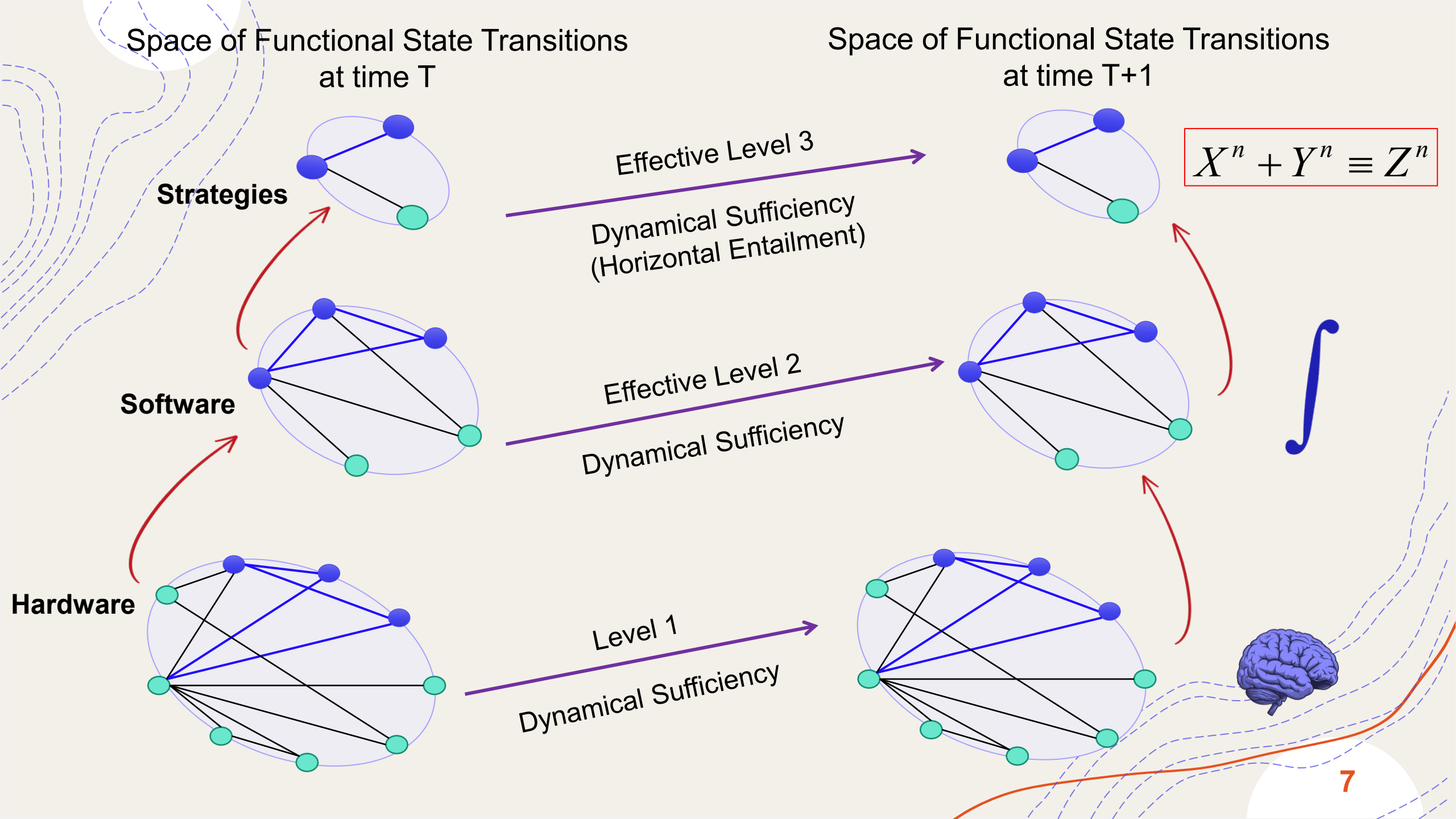


- Emergence 101

“When There is No Need to Look Under The Hood”

Space of Functional State Transitions at time T

Space of Functional State Transitions at time T+1

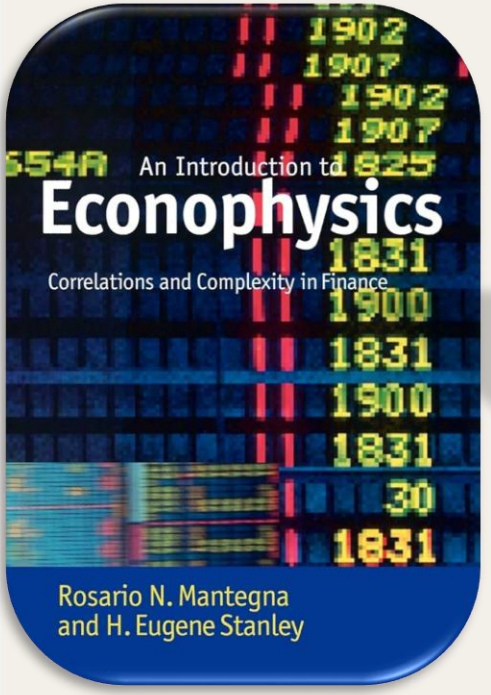




- **Complexity Explorer Lecture: David Krakauer - What is Complexity?**

https://www.youtube.com/watch?v=FBkFu1g5PlE&t=1558s

- A History of Econophysics



Physica A 179 (1991) 232–242
North-Holland

PHYSICA A

Lévy walks and enhanced diffusion in Milan stock exchange

Rosario Nunzio Mantegna

Max-Planck-Institut für Quantenoptik, W-8046 Garching, Germany[1]
and Istituto di Fisica, via Archirafi 36, I-90123 Palermo, Italy[2]

Received 7 March 1991

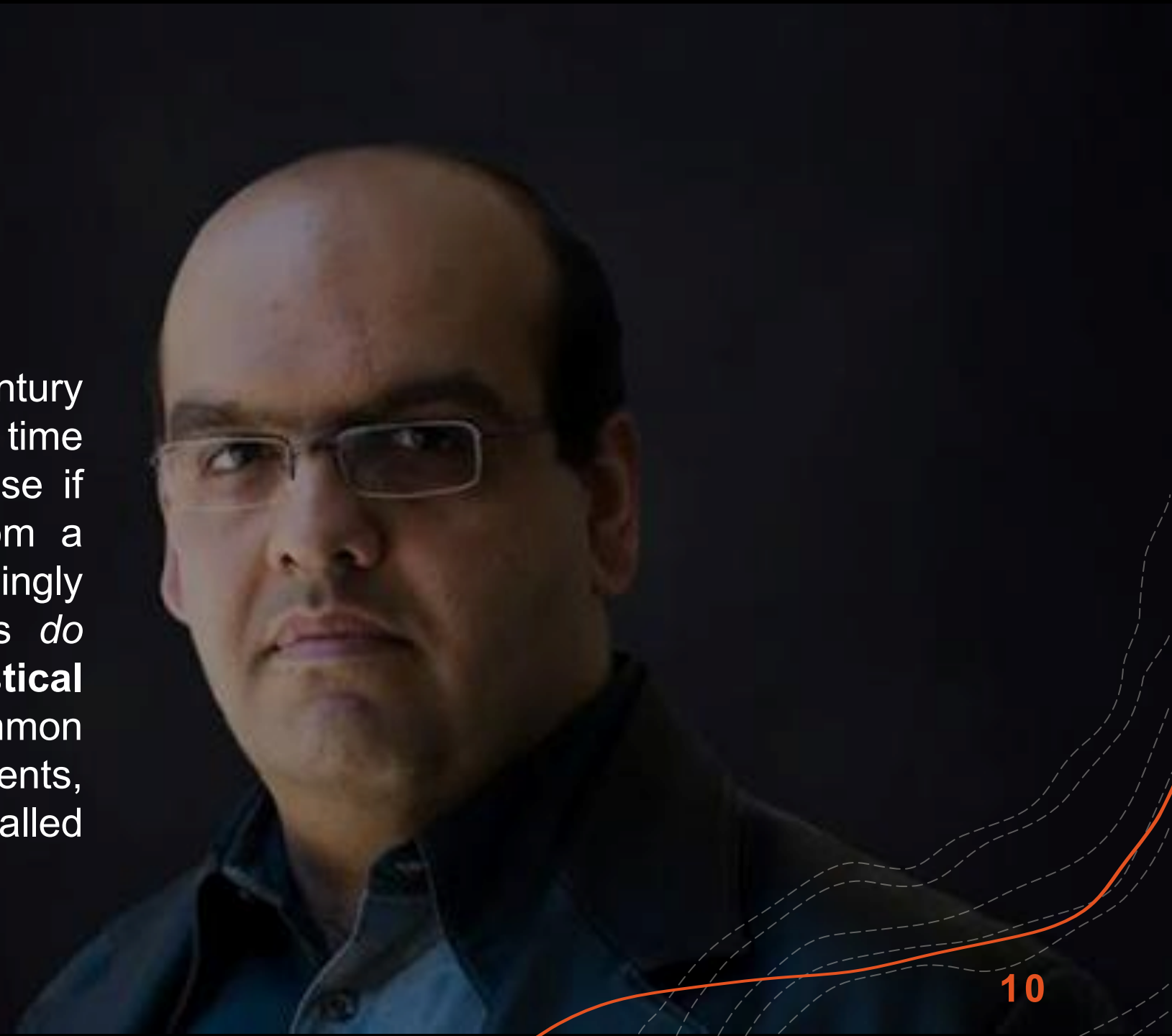
We perform statistical analyses of the general and sectorial historical M.I.B. indices of the Milan stock exchange. Our analyses show that the price indices have statistical properties which are compatible with a Lévy random walk. The time evolution of the daily variations of indices is intermittent on a time scale of years and the variance of almost all indices displays a superdiffusive behavior. By using the theory of enhanced diffusion in Lévy walks as theoretical framework we ascribe the superdiffusive behavior to a nonlocal memory coupling price and time.

Fig.3 - The very first paper in Econophysics - 1991



Prof. Rosario N. Mantegna
Palermo University

The Empirical Stylized Facts



+ “The result of more than half a century of empirical studies on financial time series indicates that this is the case if one examines their properties from a *statistical* point of view: the seemingly random variations of asset prices *do* share some **quite nontrivial statistical properties**. Such properties, common across a wide range of instruments, markets and time periods are called ***stylized empirical facts***.”

+ Rama Cont

$$r_\tau(t) = \log p(t+\tau) - \log p(t)$$

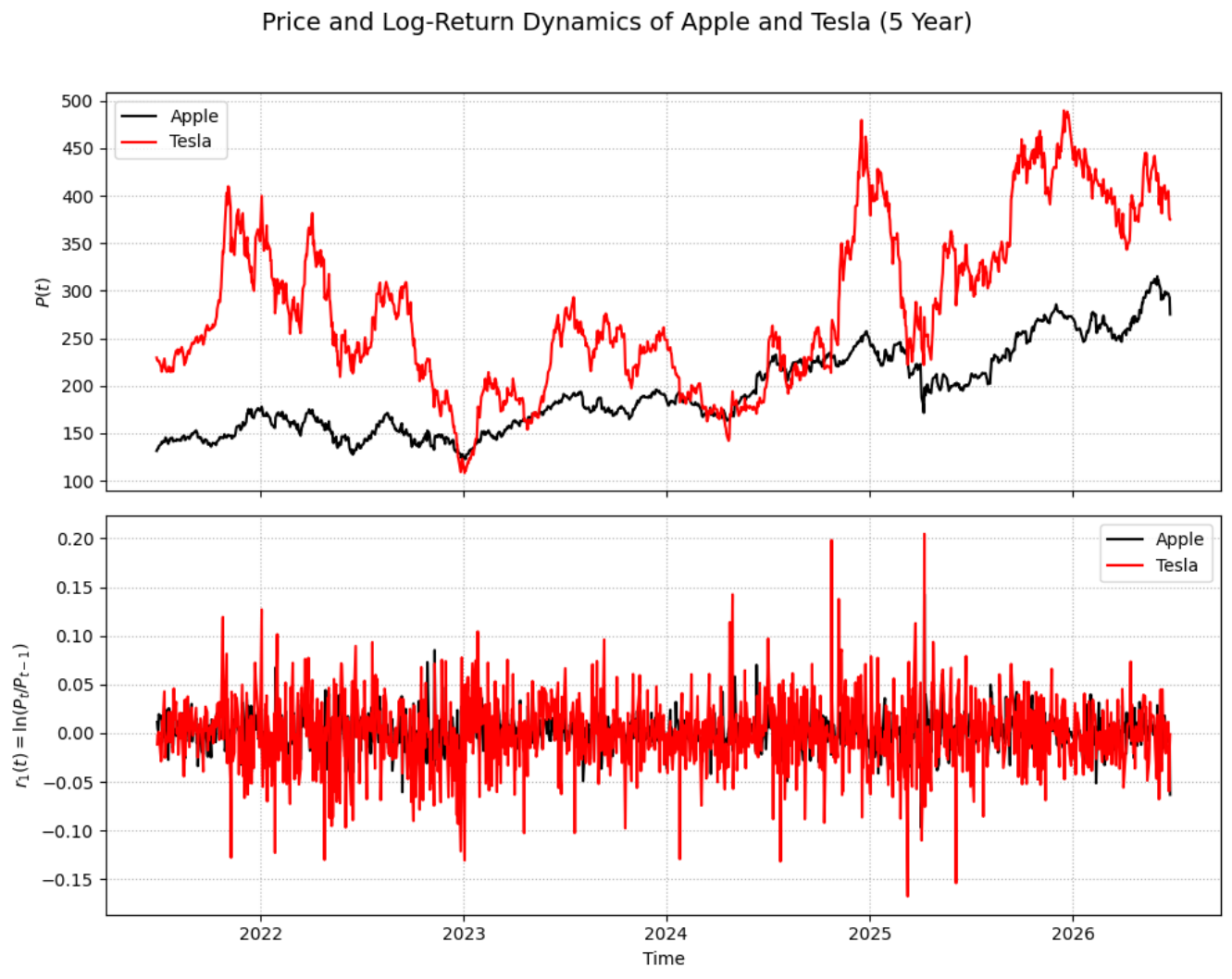


Fig.4 – Price and Log-Return Dynamics of Apple and Tesla in the last 5 Years

- Fat Tail Distribution

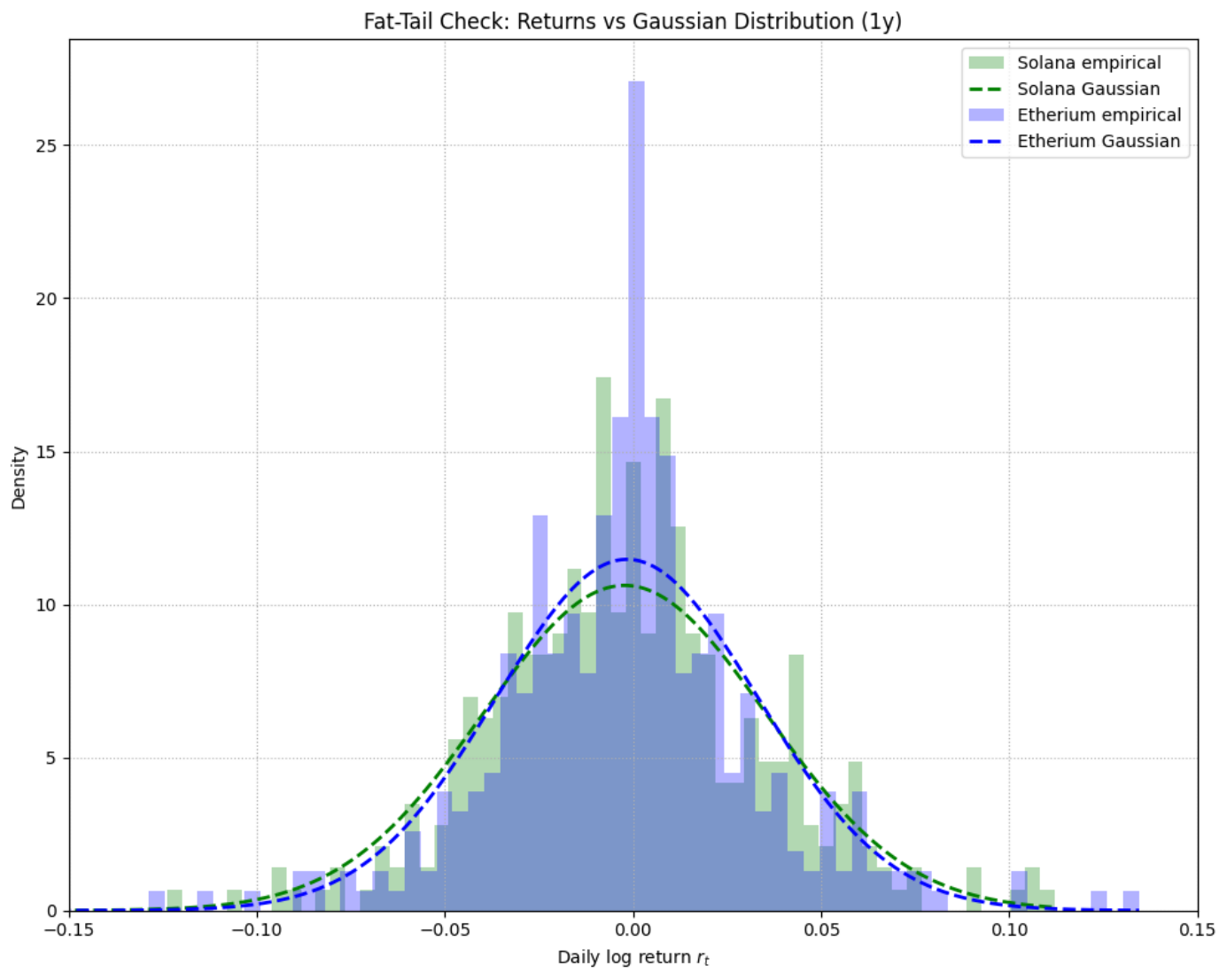


Fig.5 – Fat-Tail check for Solana and Ethereum in the last year

Table 1: Fat-Tail Statistical test

Asset	Excess kurtosis
Solana	1.778
Ethereum	2.702

$$\gamma = \frac{\langle (r_\tau(t) - \langle r_\tau(t) \rangle)^4 \rangle}{\sigma_\tau^4} - 3$$

- Aggregational Gaussianity

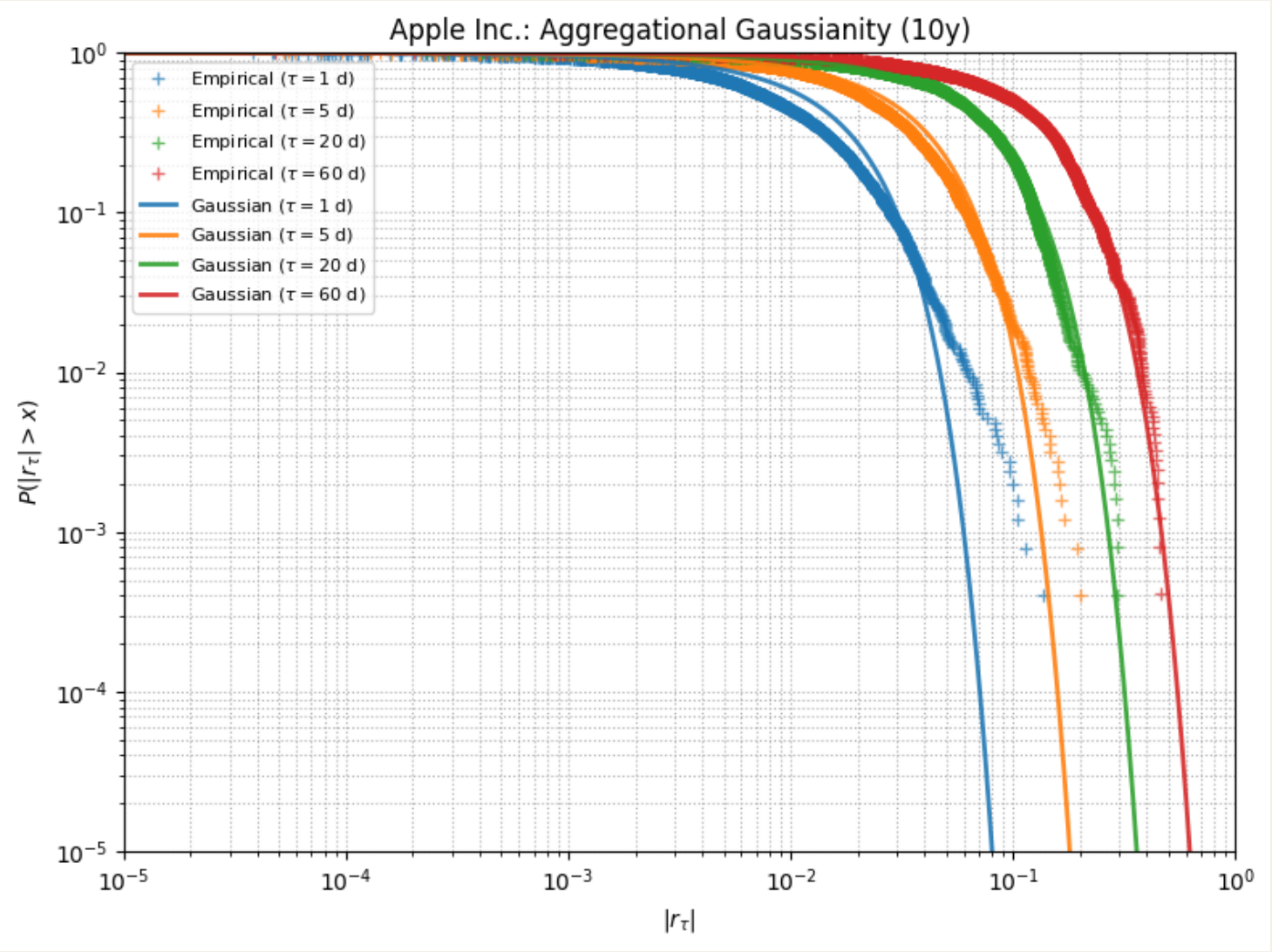


Fig.5 – Aggregational Gaussianity of Apple for last 10 years

- Absence of Autocorrelations

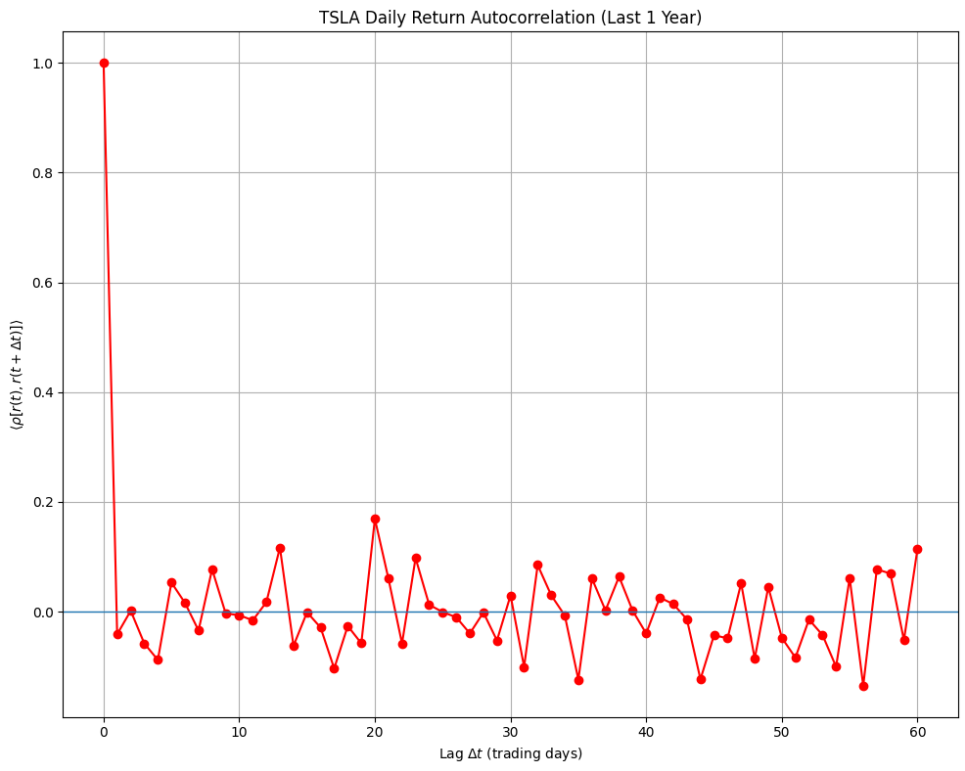


Fig.6 – Daily Return Autocorrelation of Tesla for the Last 1 year

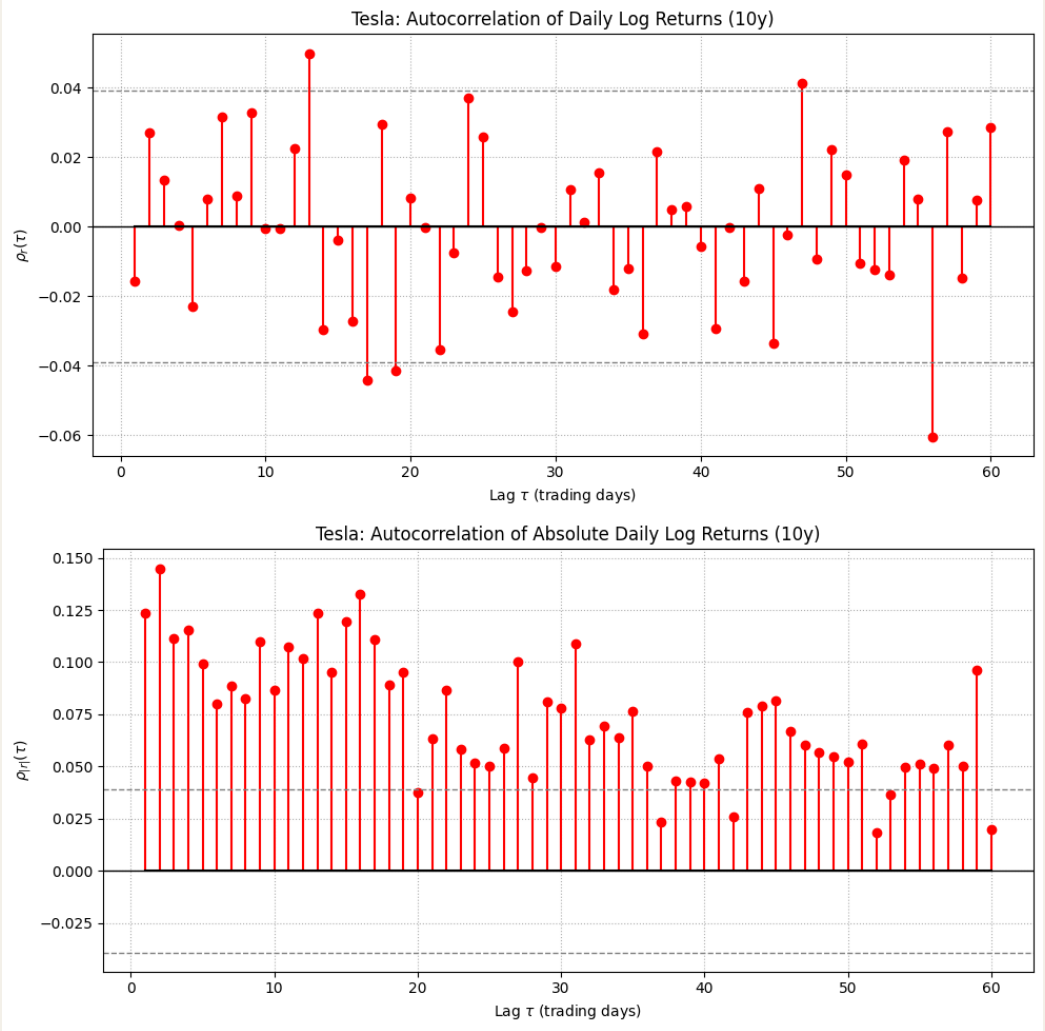


Fig.7 & 8 – Autocorrelation of Tesla for Daily/Absolute Log Returns in the last 10 years



- "The efficient market hypothesis (**EMH**) implies the unpredictability of future returns and the absence of bubbles that can be exploited for arbitrage opportunity."

Nobel Memorial Prize in Economic Sciences – 2013





- Eugene Fama (father of modern finance) - University of Chicago Booth School of Business

- Robert J. Shiller - Yale University

I. Markets are **highly efficient** and quickly reflect available information.
II. Emphasizes rational pricing.

I. Markets are **not perfectly efficient** because investors exhibit psychological biases.
II. Emphasizes investor sentiment and irrational exuberance.



How market intervention can prevent bubbles and crashes

Rebecca Westphal[a,d], Didier Sornette[a,b,c,e]

[a]ETH Zürich, Department of Management, Technology and Economics,
Scheuchzerstrasse 7, 8092 Zürich, Switzerland,
[b]Institute of Risk Analysis, Prediction and Management, Academy for Advanced Interdisciplinary Studies,
Southern University of Science and Technology (SUSTech), Shenzhen, 518055, China
[c]Institute of Innovative Research, Tokyo Tech World Research Hub Initiative (WRHI)
Tokyo Institute of Technology, Tokyo, Japan
[d]E-mail address: rwestphal@ethz.ch
[e]E-mail address: dsornette@ethz.ch

August 29, 2020

Fig.9 – “How market intervention can prevent bubbles and crashes” paper

- Volatility Clustering



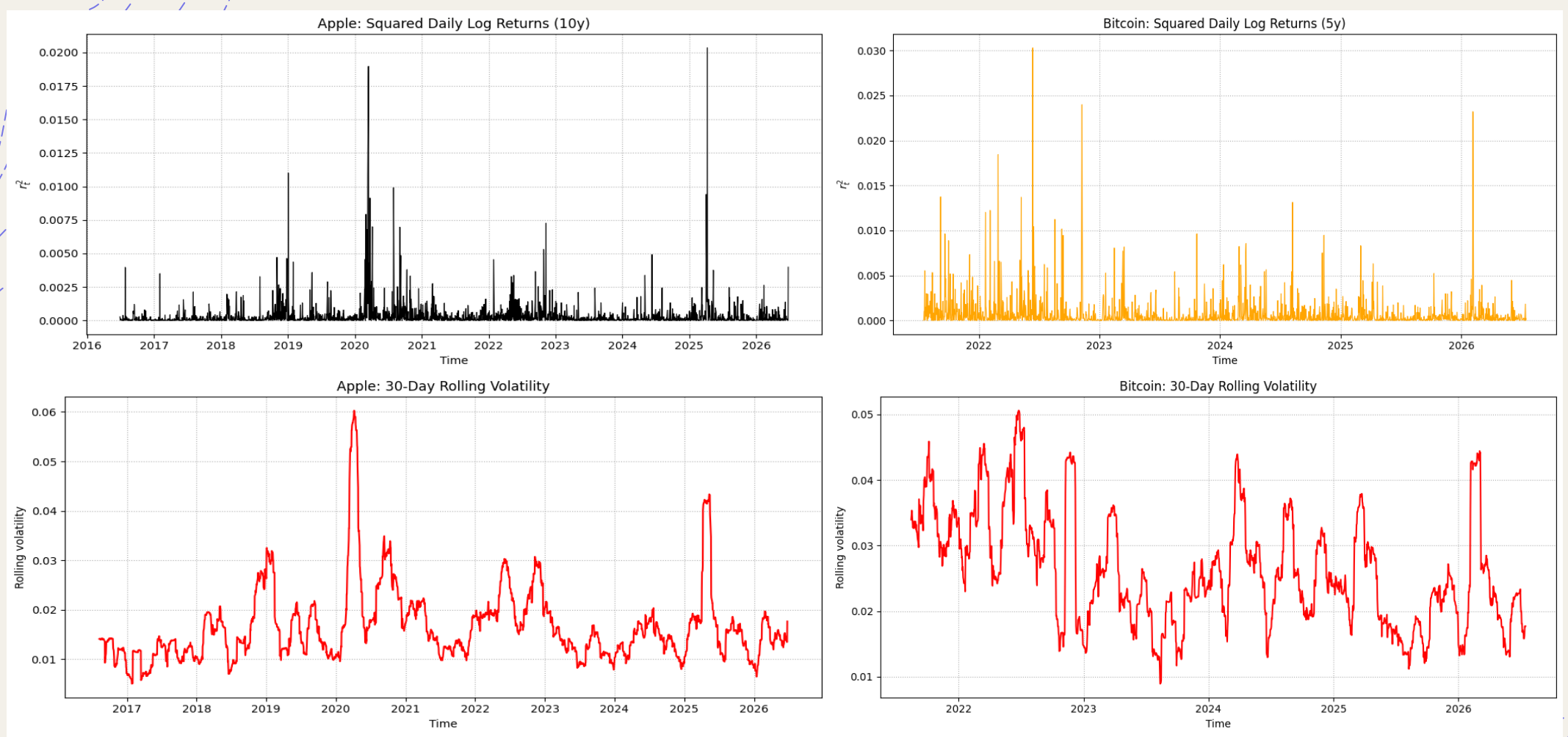


Fig.10 – Squared Daily Log Returns and 30-Day Rolling Volatility for Apple and Bitcoin

- Hurst Exponent

For a Stationary Standard Gaussian Function $X(t)$ we have $\begin{matrix}\mathbb{E}[X(t)] = 0 \\ \mathbb{E}[X^2(t)] = 1\end{matrix}$ so the Autocorrelation function $C(\Delta t) = \mathbb{E}\{X(t)X(t+\Delta t)]$ Could be useful to study profile roughness of the process if it behaves like:

$$C(\Delta t) \sim 1 - |\Delta t|^{\alpha}, \quad for \quad \Delta t \to 0$$

And for its fractal dimension we can say:

$$d_F = 2 - \frac{\alpha}{2} \quad for \quad \alpha \in (0, 2],$$

By analyzing asymptotic behavior at large time delays, for its long-term dependency, we can say:

$$C(\Delta t) \sim |\Delta t|^{-\beta}, \quad for \quad \Delta t \to +\infty$$

Hurst Exponent will be:

$$H = 1 - \frac{\beta}{2} \quad for \quad \beta \in (0,1)$$

For Self-Similar process in an n-dimensional space, local properties would preserved in larger scales so we have:

$$d_F = n + 1 - H$$

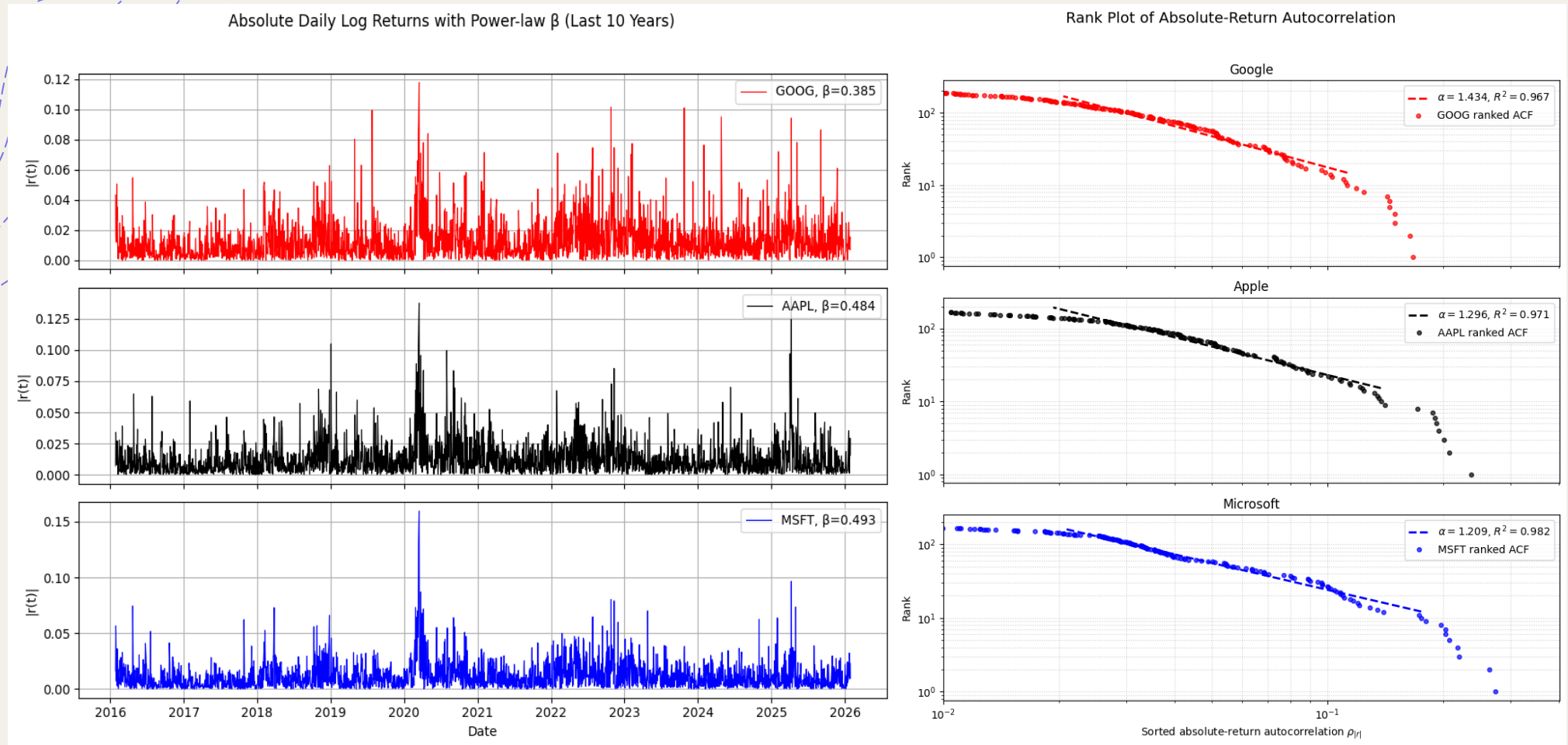


Fig. 11 – Absolute Daily Log Returns of Google, Apple, and Microsoft for last 10 years

- R/S Analysis

For Time series $X(t)$ when $t = n, 2n, 3n, \ldots$ We have Mean and Standard deviation like:

$$\langle X \rangle_T = \frac{n}{T} \sum_{K=1}^{T/n} X(kn) \qquad S(T) = \left(\frac{n}{T} \sum_{K=1}^{T/n} [X(kn) - \langle X \rangle_T]^2 \right)^{1/2}$$

The R-range of such time series is defined as the difference between the minimum and maximum values of the time series within the interval $[n, T]$:

$$R(T) = \max_{n \le t \le T} [X(t)] - \min_{n \le t \le T} [X(t)]$$

Then we can define Hurst Exponent using the scaling relation below:

$$\frac{R(T)}{S(T)} \sim \left(\frac{T}{n} \right)^H$$

- Generalized Hurst Exponent (GHE):



$$\Xi(\tau, q) = \mathbb{E}[|X(t+\tau) - X(t)|^q] \sim K_q \tau^{qH_q}$$

Computes the q-th order moments of increments over different time scales, assuming a power-law scaling behavior.

$$\ln\big(\Xi(\tau, q)\big) = qH_q \ln(\tau) + \ln\big(K_q\big)$$

Taking logarithms transforms the scaling law into a linear relationship, allowing H_q to be estimated from the slope.

$$qH_q = Aq + Bq^2$$

The dependence of H_q on q characterizes multifractality; a nonzero quadratic coefficient $(B \neq 0)$ indicates multifractal scaling, whereas $B = 0$ corresponds to a monofractal process.

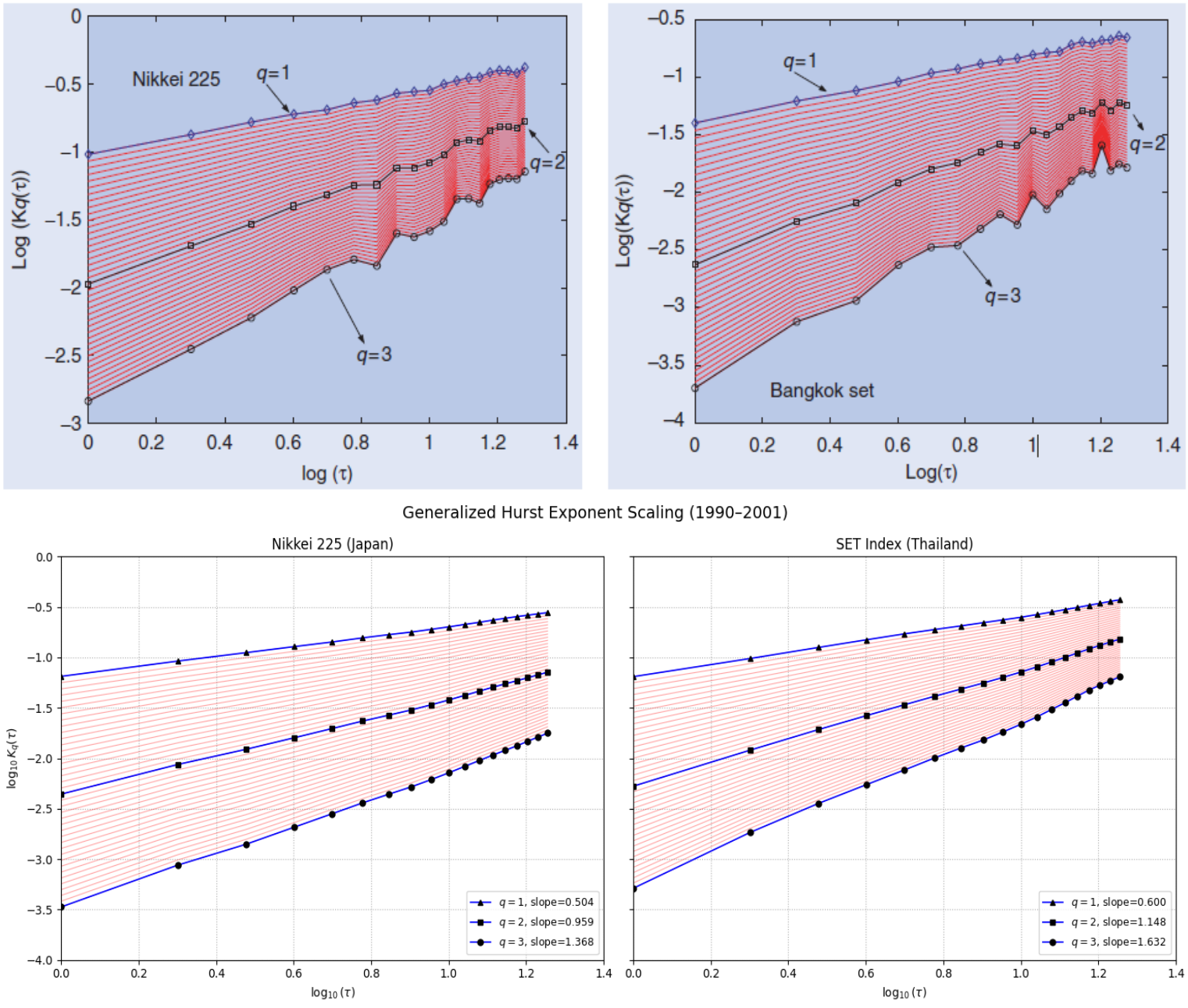
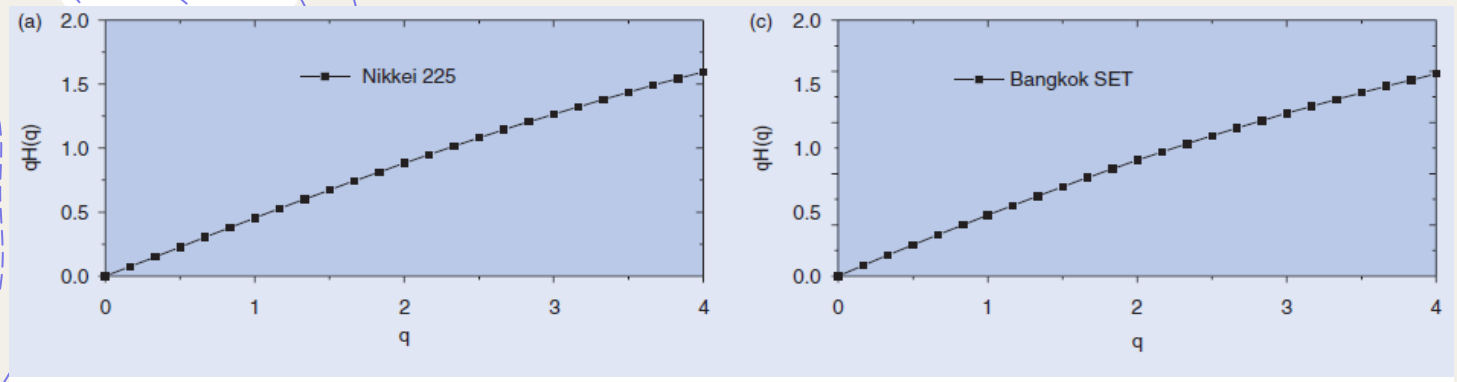


Fig. 12– GHE Analysis of the stock index of Japan and Thailand from 1990 to 2001 (Literature Review and Extracted plot)



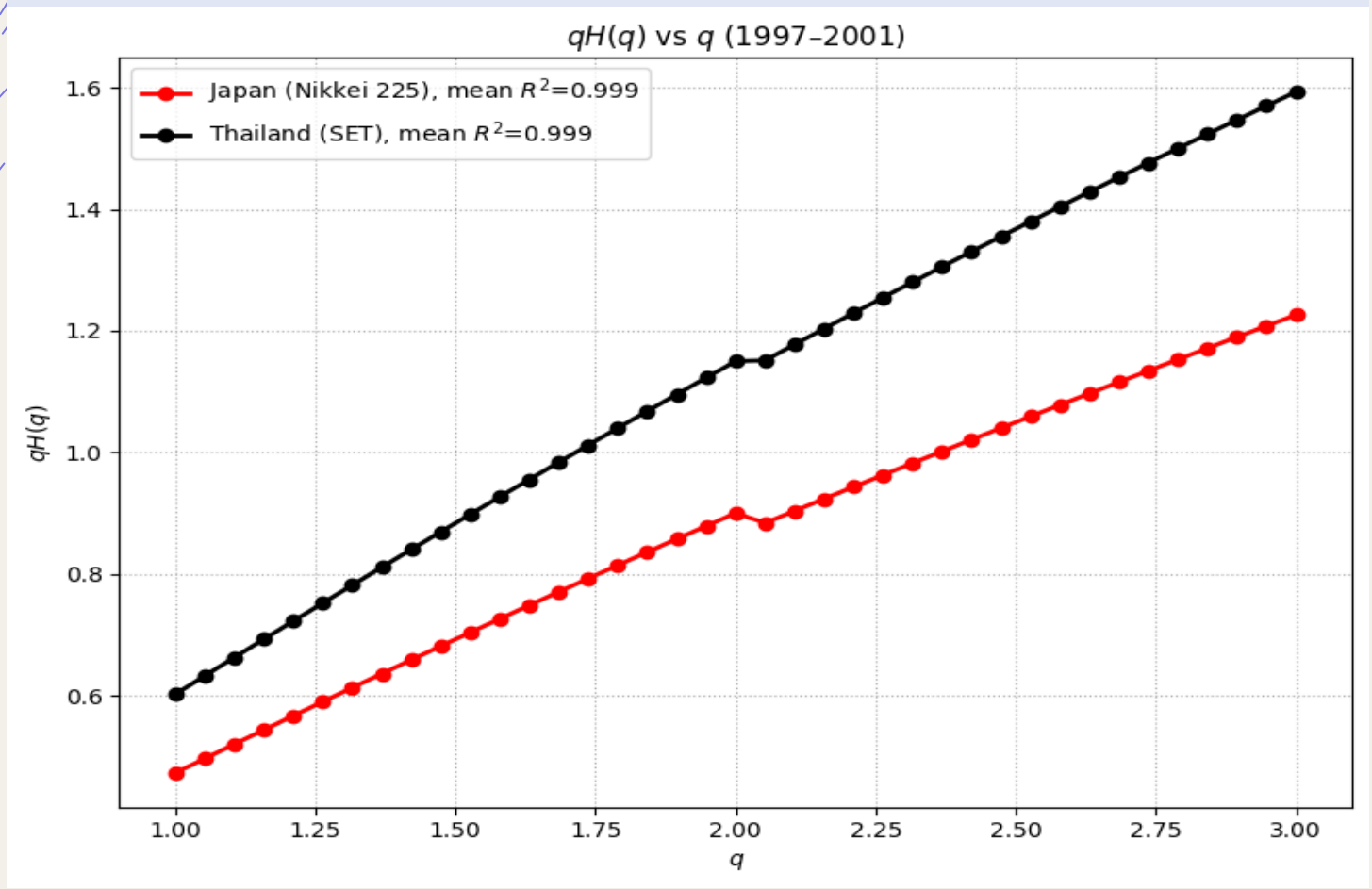


Fig. 13 - qH(q) vs q plot of the stock index of Japan and Thailand from 1997 to 2001 (Literature Review and Extracted plot)

$$S(f) \propto f^{-2H(2)-1}$$

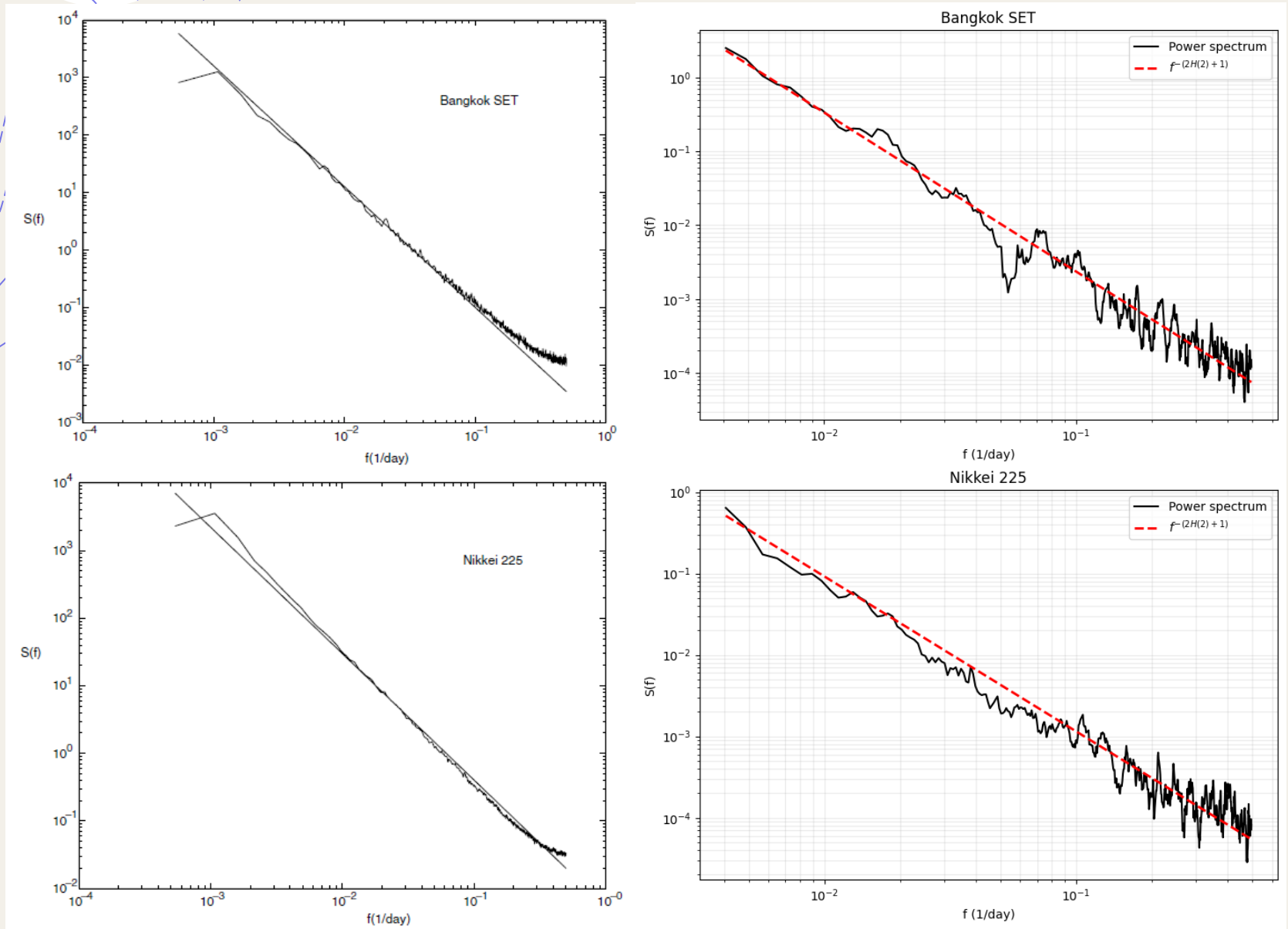
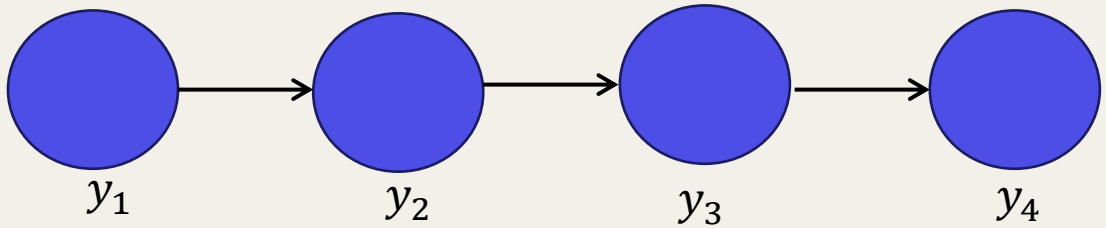


Fig. 14 - Scaling spectral density and Hurst exponent Analysis of the stock index of Japan and Thailand from 1997 to 2001 (Literature Review and Extracted plot)

- **Markov Models**



I. Autoregressive Processes
II. Moving Averages

→ ARMA (p,q) or ARIMA (p,d,q)


OMP
PIRELLI
IWC
7

- **ARCH/GARCH Models**

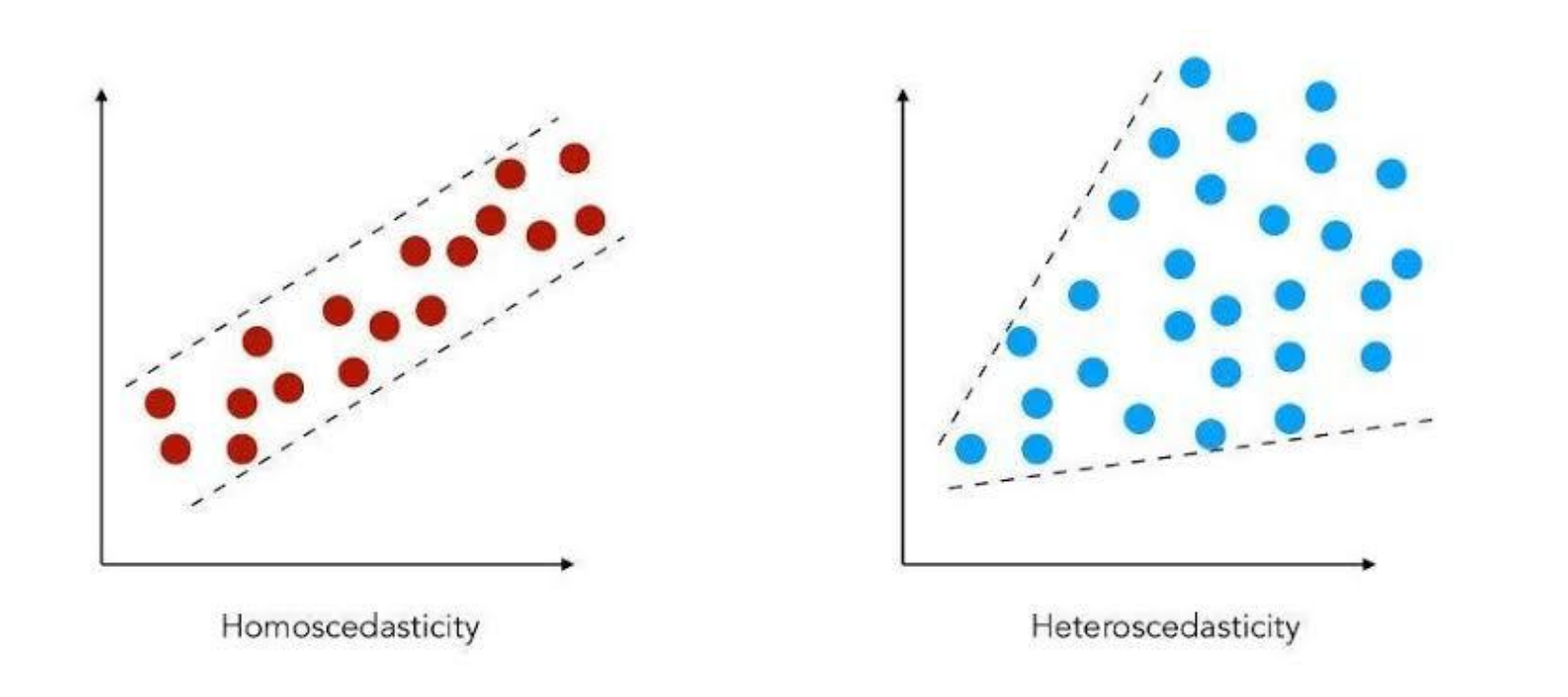


Fig. 15 – Homoscedasticity Vs Heteroscedasticity

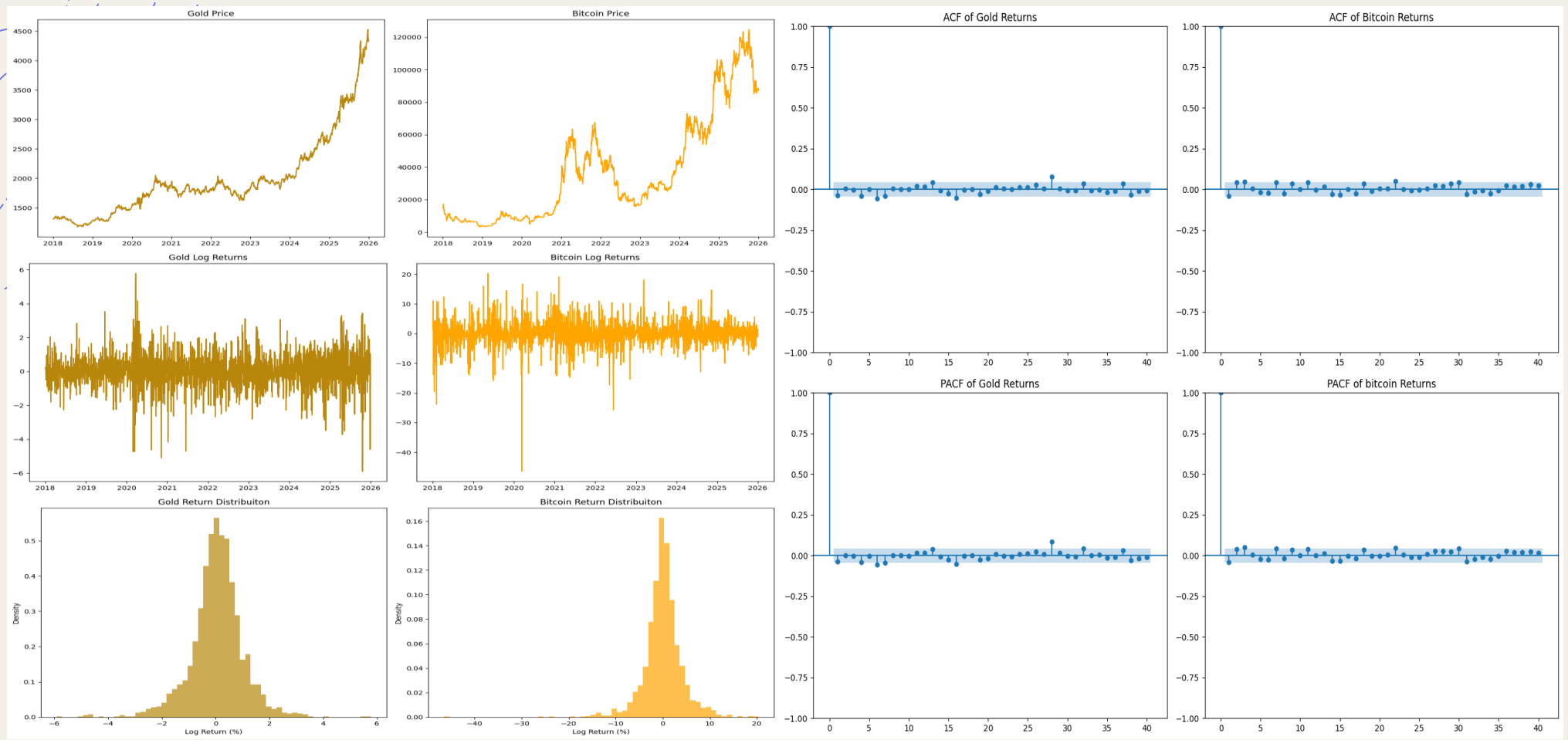


Fig. 16 – Data Analysis before linear modeling on financial data

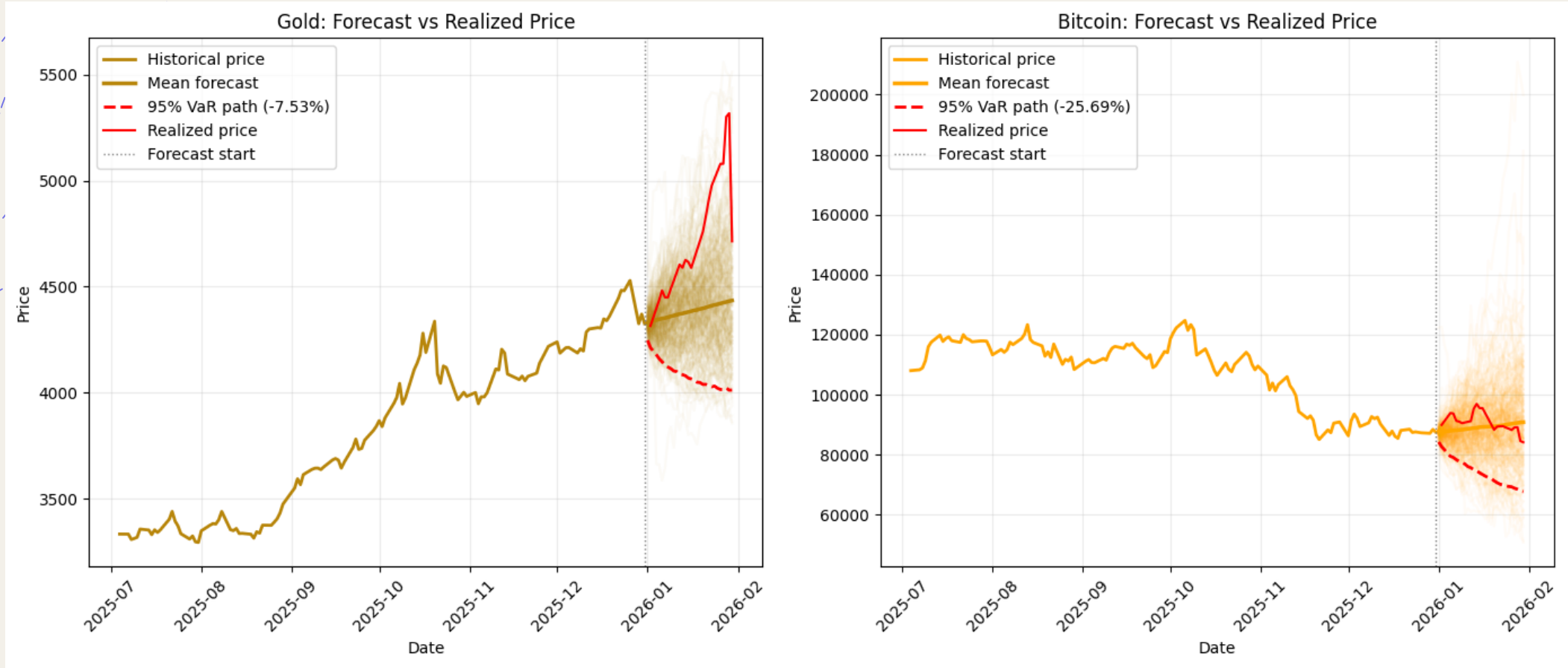


Fig. 17 – Forecasting future prices using past values for Bold and Bitcoin

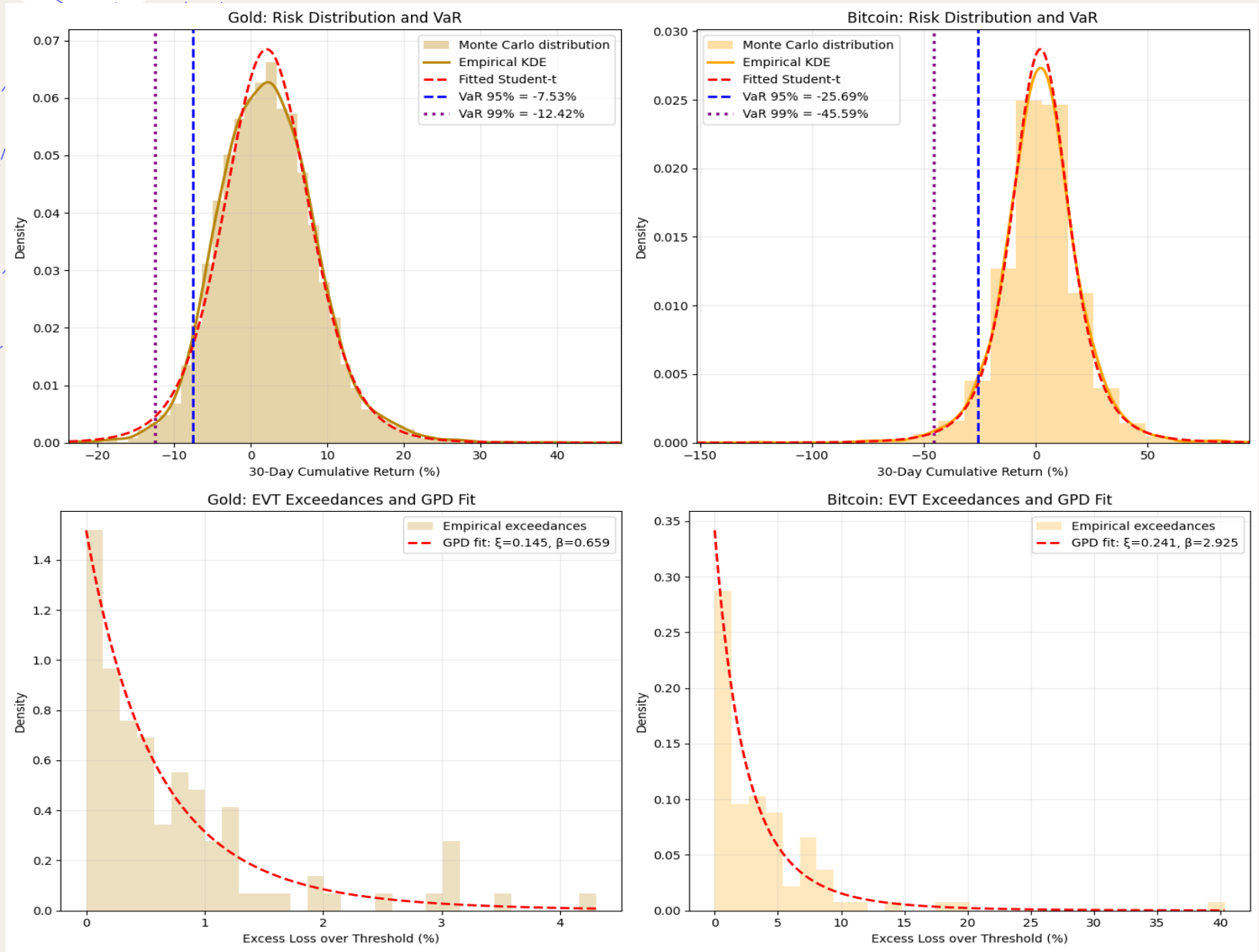


Fig. 18 – Risk and Extreme value analysis

Sequence modeling

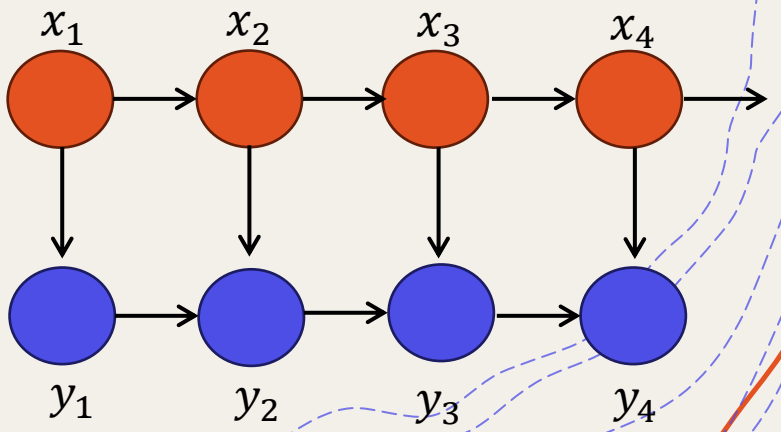
Build a probabilistic model for the observable signal $y = \{y^{(t)}\}_{t=1}^{T}$

$$p\left(y^{(1)}, y^{(2)}, \ldots, y^{(T)}\right) = \prod_{n=1}^{T} p\left(y^{(n)} \middle| y^{(1)}, y^{(2)}, \ldots, y^{(n-1)}\right)$$

Possible modeling approaches for $p(y^{(t)} | y^{(t-1)}, y^{(t-2)}, \ldots)$:

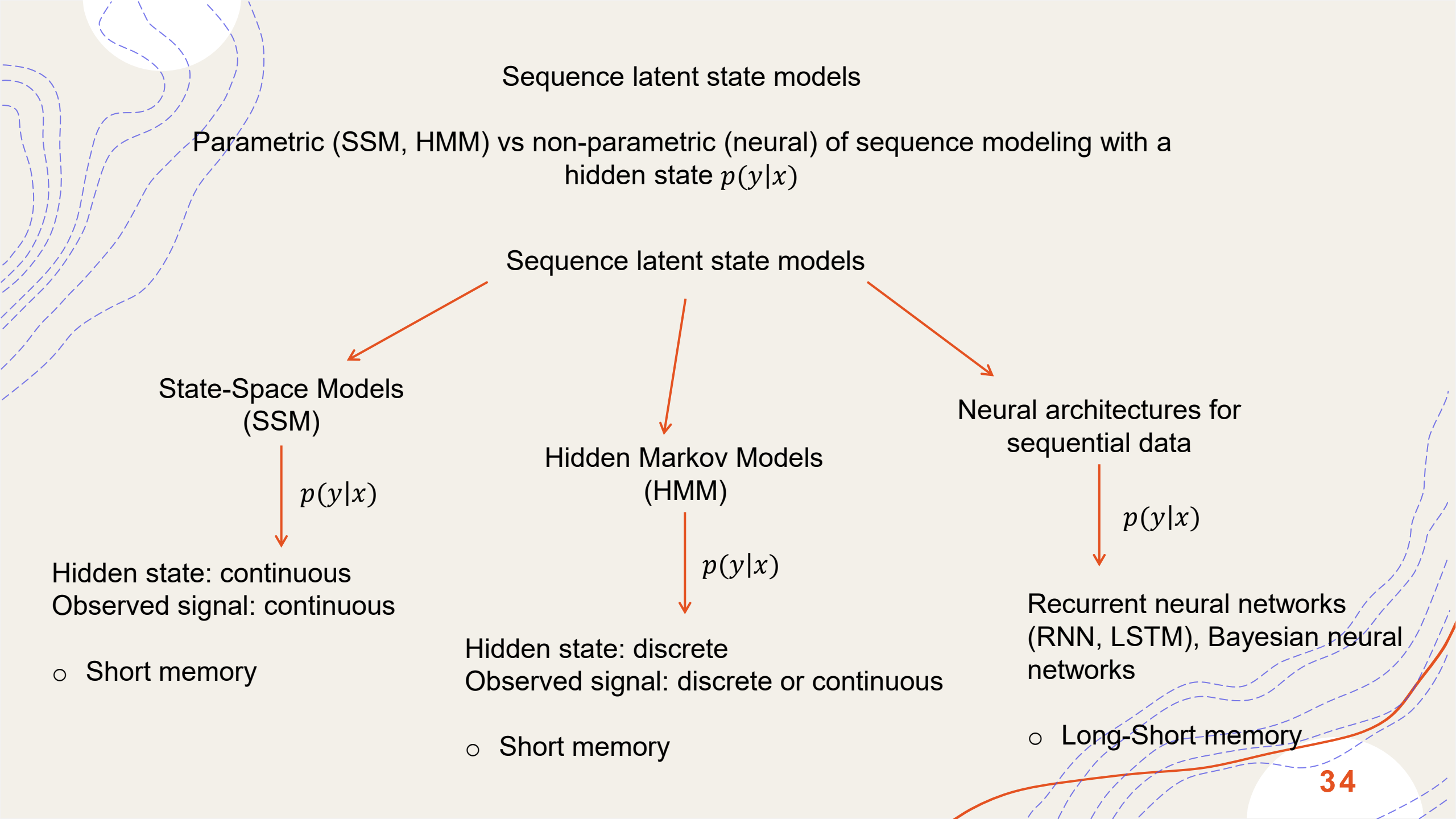
1. Markov models:
2. Dynamic hidden (latent) variables models, with an unobservable state $x^{(t)}$

- The hidden state $x^{(t)}$ captures the dynamics of the system, filters noise out
- Used as a conditioning variable for predictions $p(y^{(t)} | y^{(t-1)})$ $\rightarrow p\left(y^{(t)} \middle| x^{(t)}\right)$.

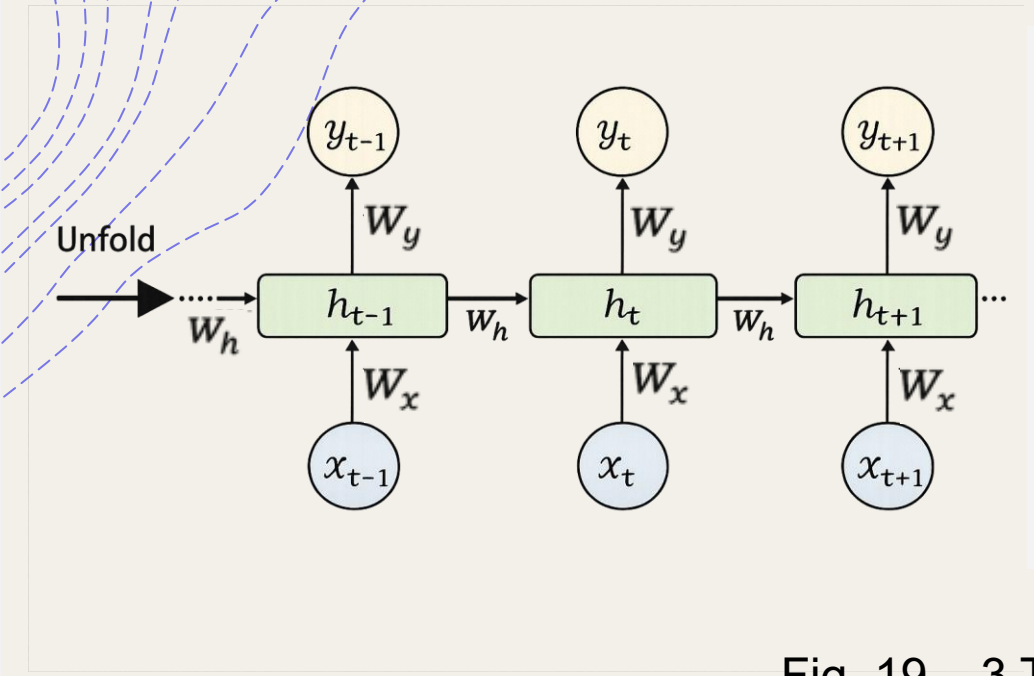


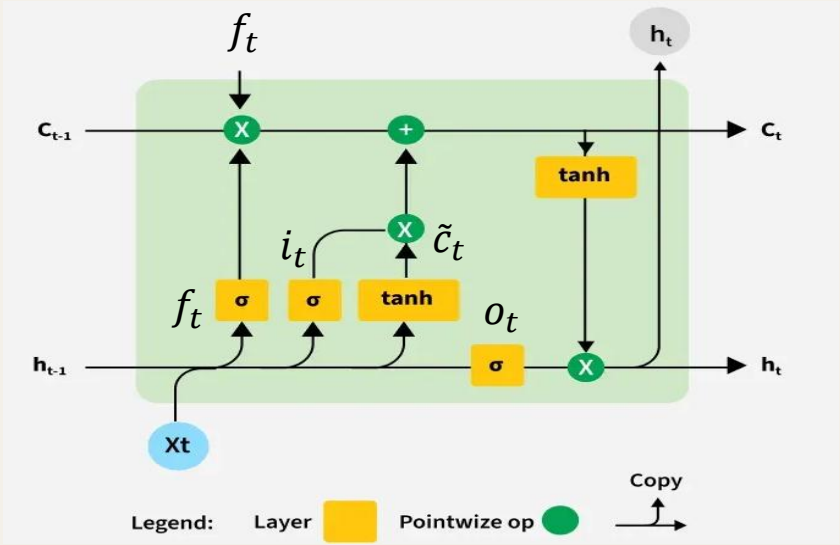
Sequence latent state models

Parametric (SSM, HMM) vs non-parametric (neural) of sequence modeling with a hidden state $p(y|x)$



- RNN, LSTM, GRU





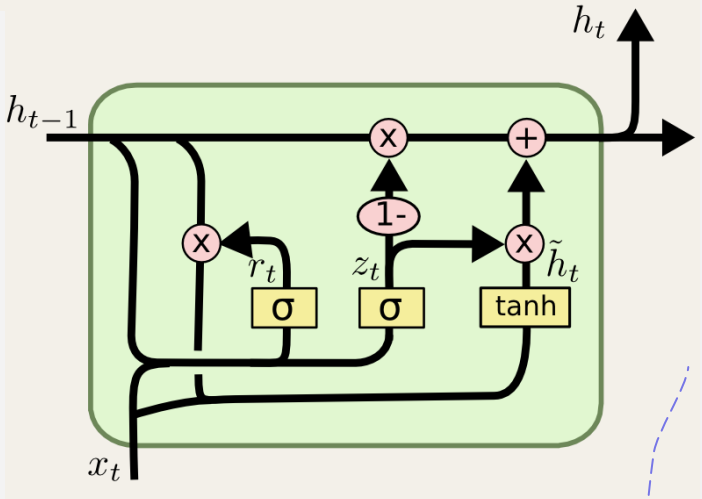


Fig. 19 – 3 Types of Recurrent neural networks

- Temporal Convolutional Networks (TCN)

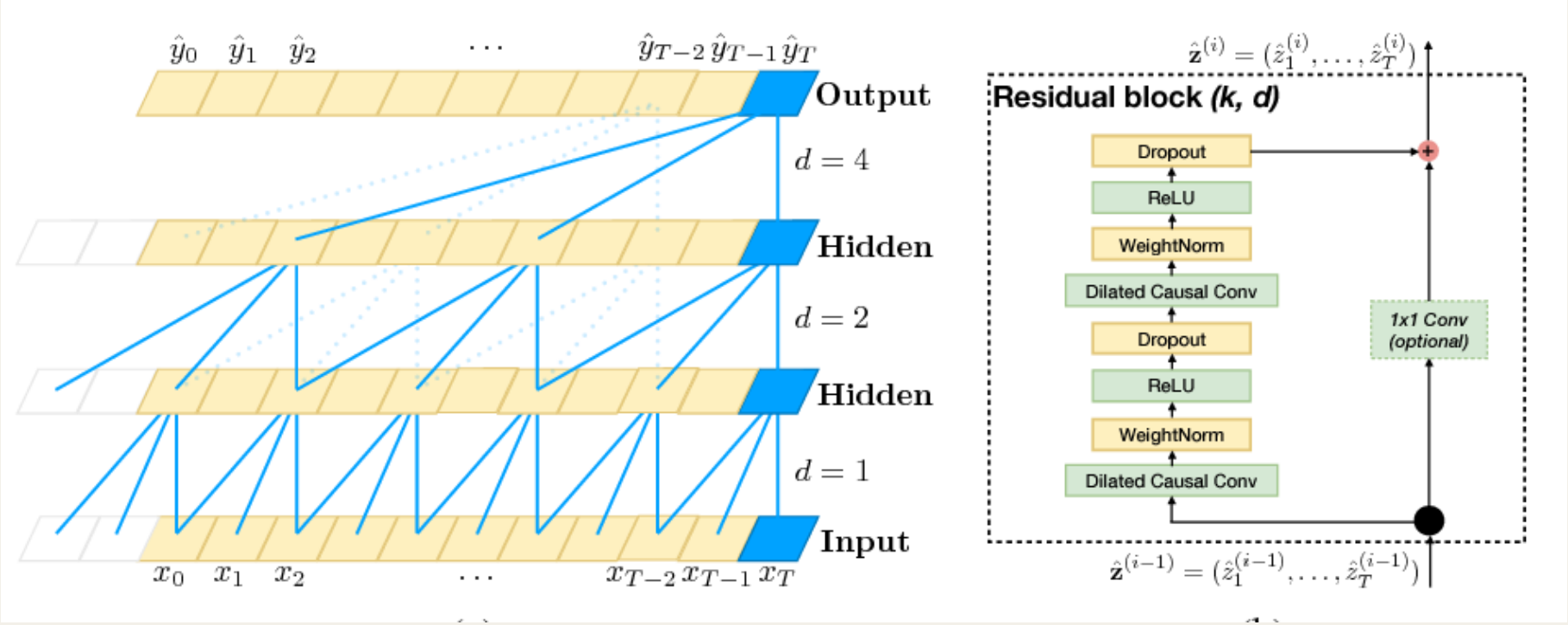
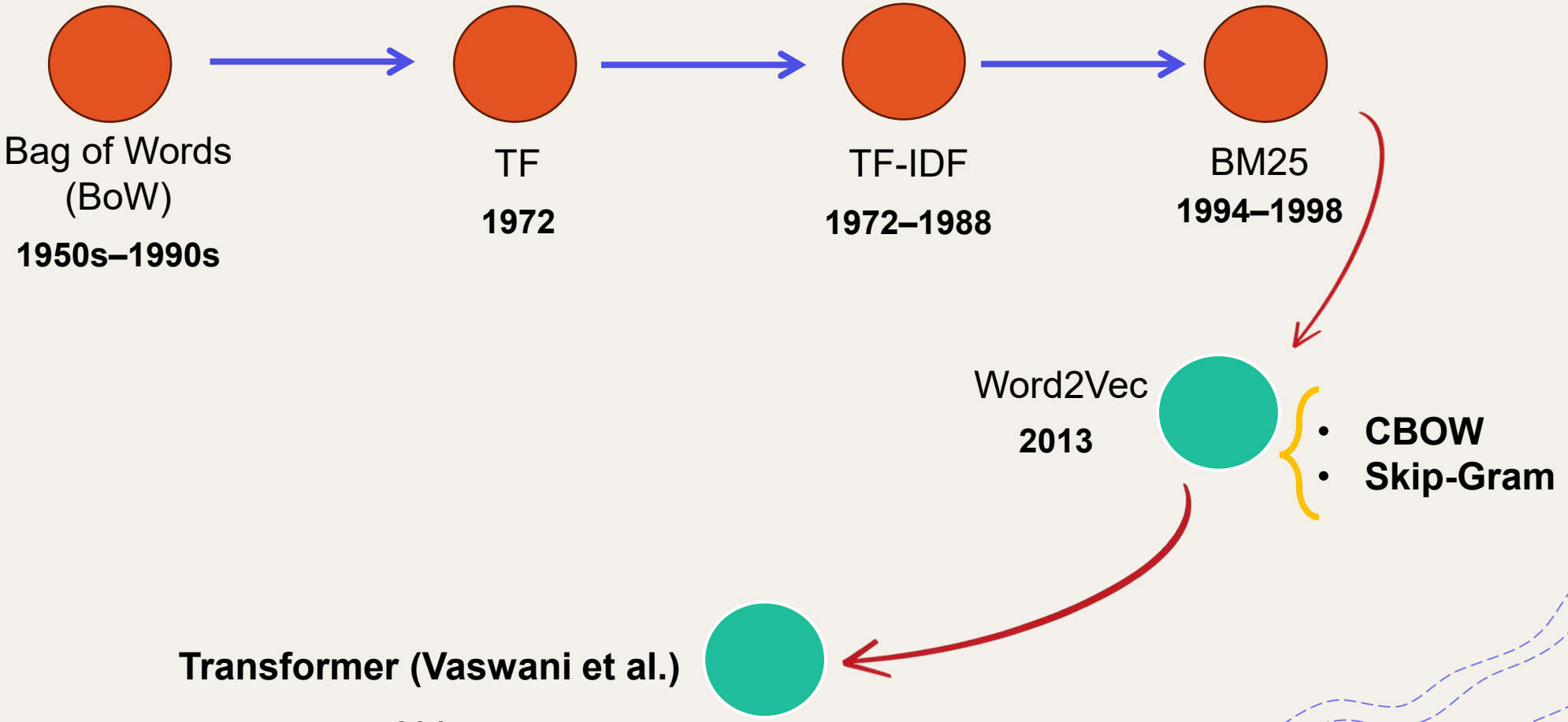
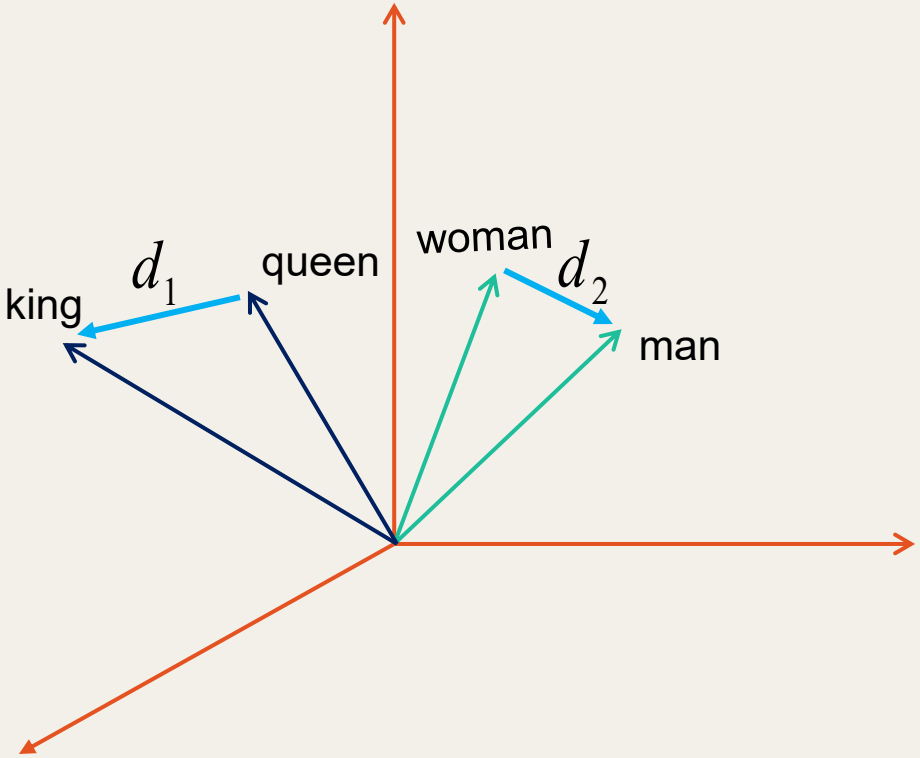


Fig. 20 – TCN Architecture

- A History of Natural Language Processing (NLP)



- Attention Mechanism



…… jumps over the fence.

- ❑ Bee
- ❑ Fox
- ❑ Rabbit
- ❑ Viper

……	**jumps**	over	the	fence

- Positional Encoding

$$PE_{(pos,2i)} = \sin(pos / 1000^{2i/d_{\text{model}}})$$

$$PE_{(pos,2i+1)} = \cos(pos / 1000^{2i/d_{\text{model}}})$$

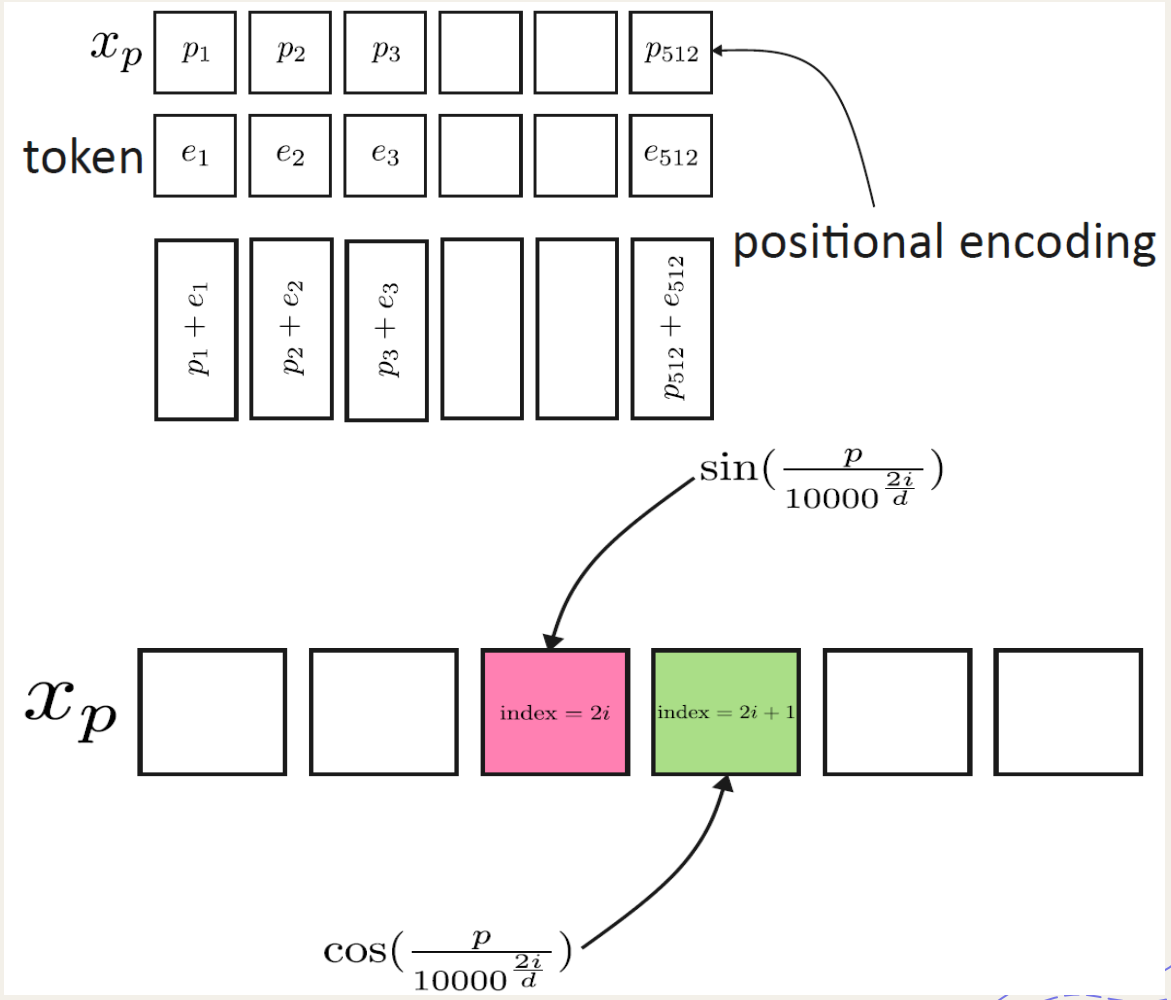


Fig. 21 – Positional Encoding

- Transformers

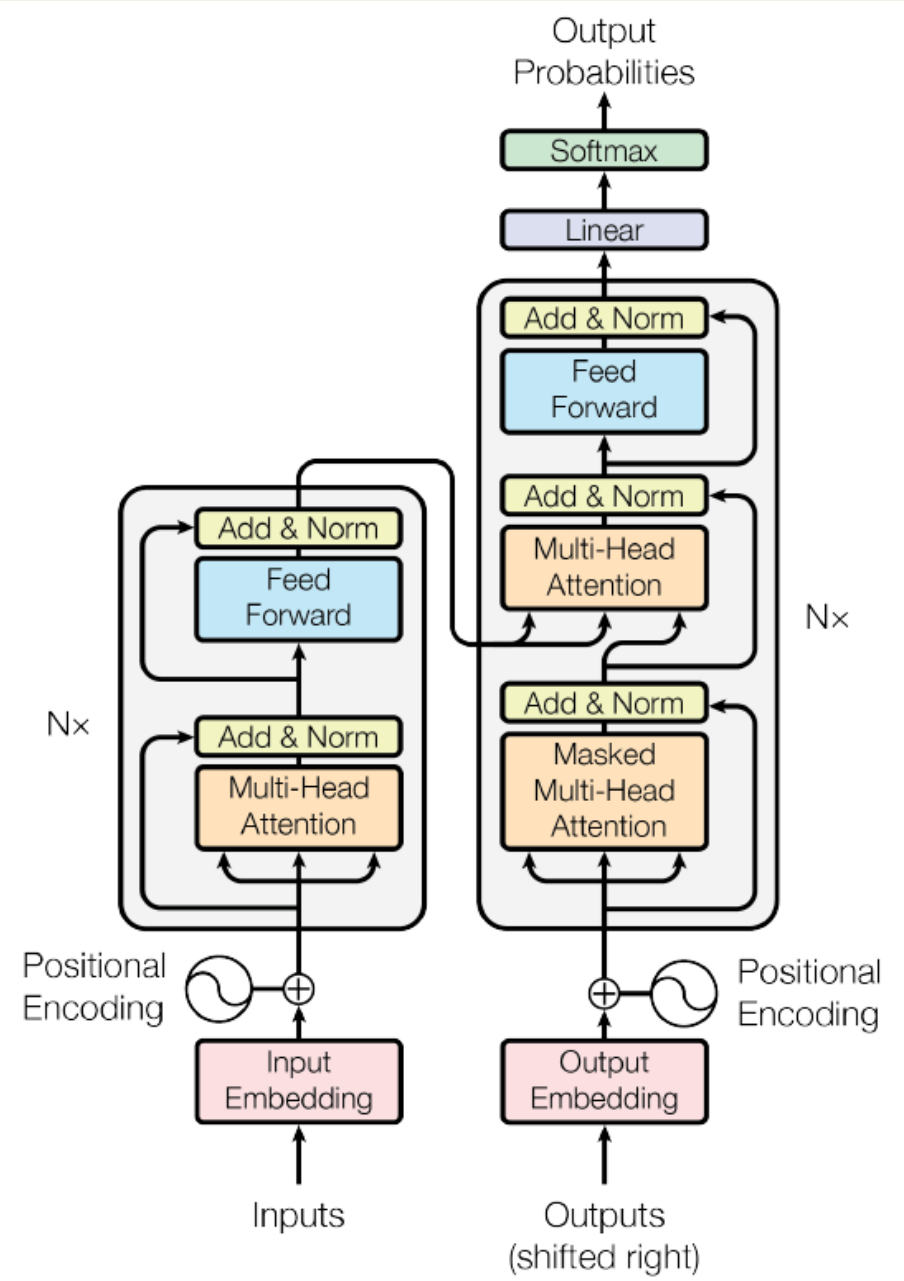
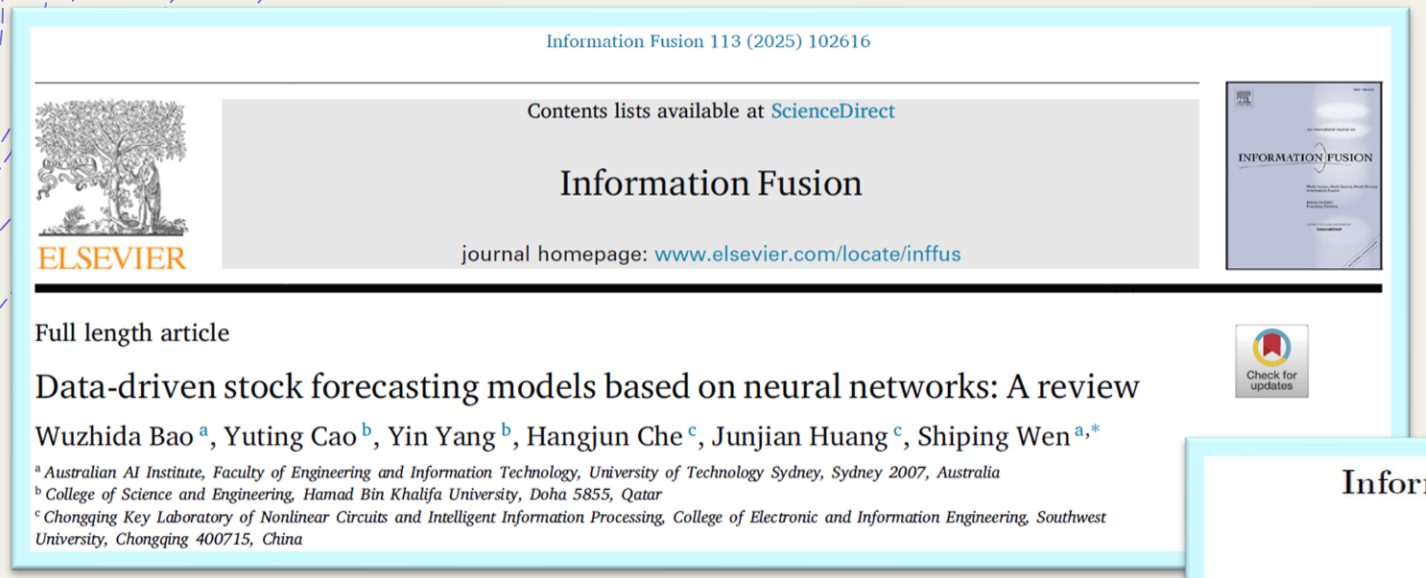


Fig. 22 – Transformer Architecture

- **Simulations and Results**



Information Fusion 113 (2025) 102616

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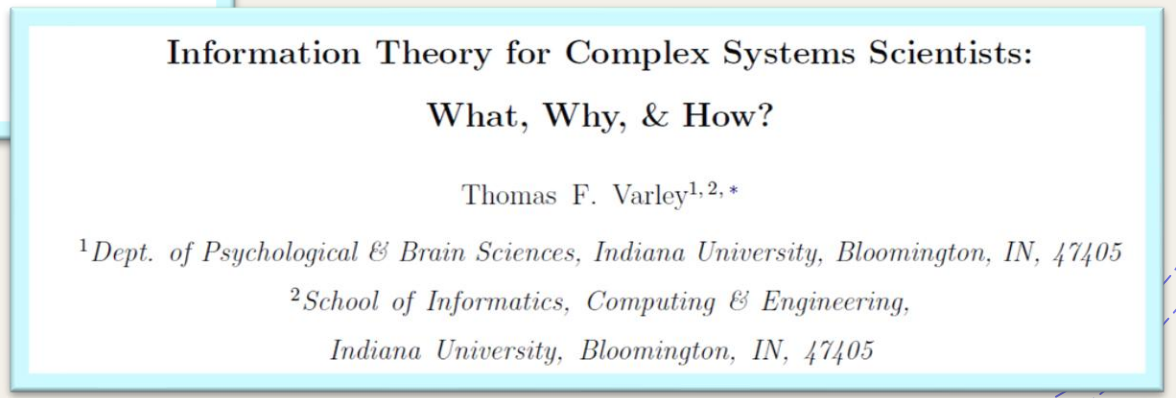
ELSEVIER

Full length article

Data-driven stock forecasting models based on neural networks: A review

Wuzhida Bao [a], Yuting Cao [b], Yin Yang [b], Hangjun Che [c], Junjian Huang [c], Shiping Wen [a,*]

[a] Australian AI Institute, Faculty of Engineering and Information Technology, University of Technology Sydney, Sydney 2007, Australia
[b] College of Science and Engineering, Hamad Bin Khalifa University, Doha 5855, Qatar
[c] Chongqing Key Laboratory of Nonlinear Circuits and Intelligent Information Processing, College of Electronic and Information Engineering, Southwest University, Chongqing 400715, China



Information Theory for Complex Systems Scientists:

What, Why, & How?

Thomas F. Varley[1,2,*]

[1] Dept. of Psychological & Brain Sciences, Indiana University, Bloomington, IN, 47405
[2] School of Informatics, Computing & Engineering, Indiana University, Bloomington, IN, 47405

Fig. 23 – Inspirational Papers for our Simulations

- First Experiment

$$dx = \mu x_t dt + g x_t dw$$

$$\frac{dx}{x} = \mu dt + g dw$$

$$r_n = \frac{x_{n+1} - x_n}{x_n} = \mu dt + g dw$$

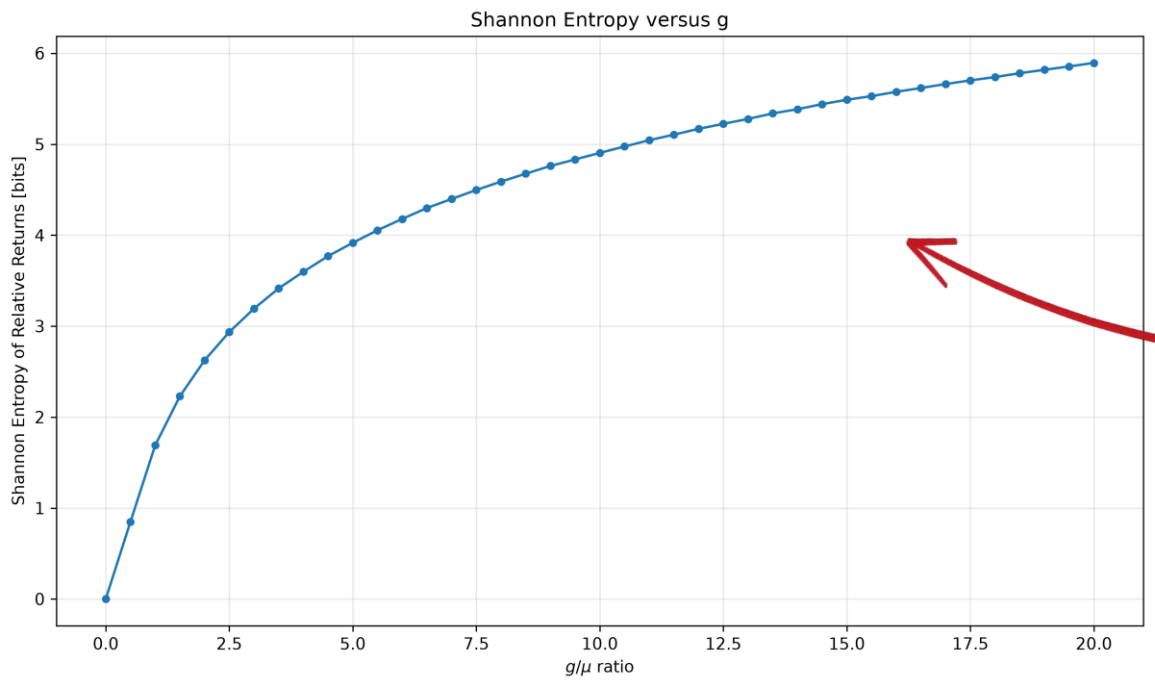


Fig. 25 – Shannon Entropy versus g

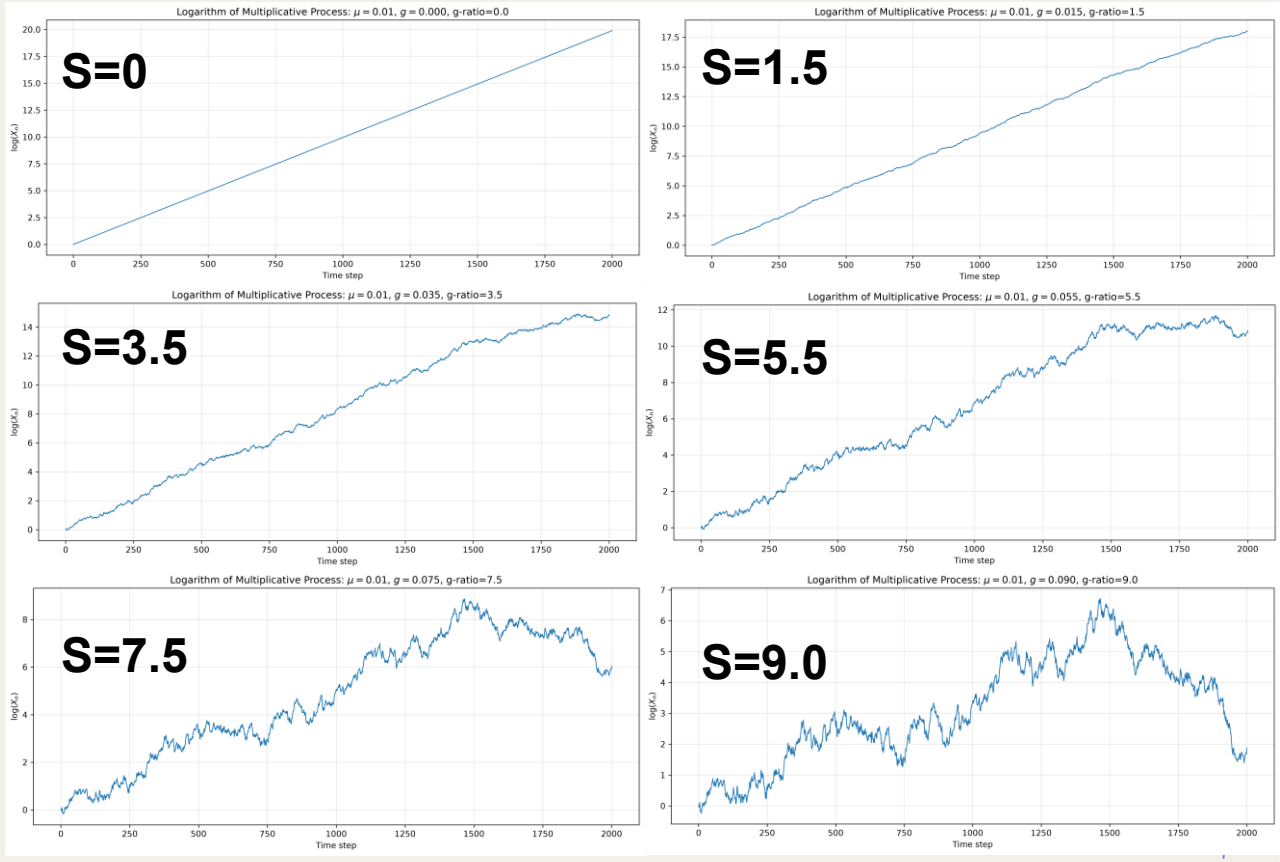
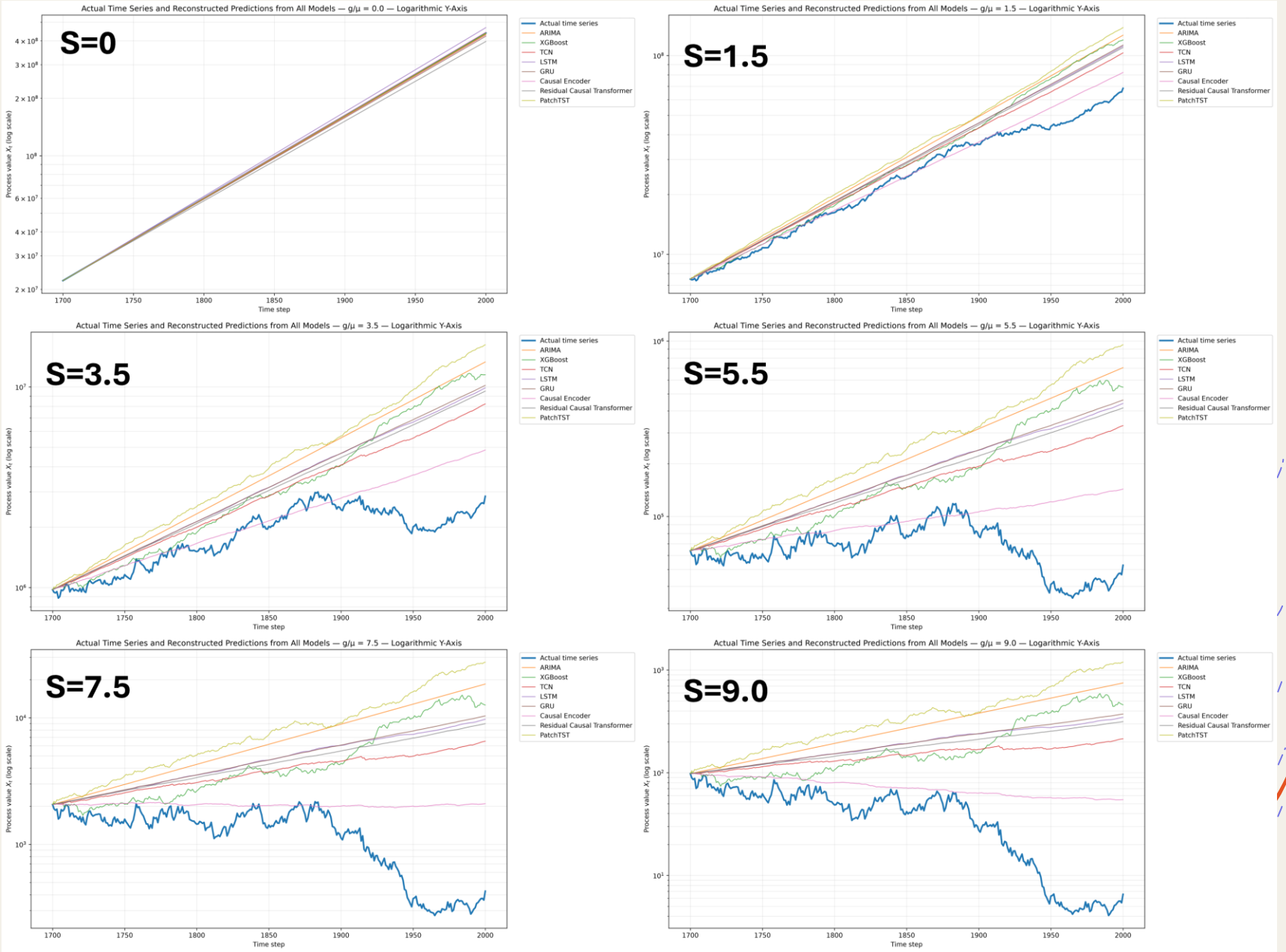


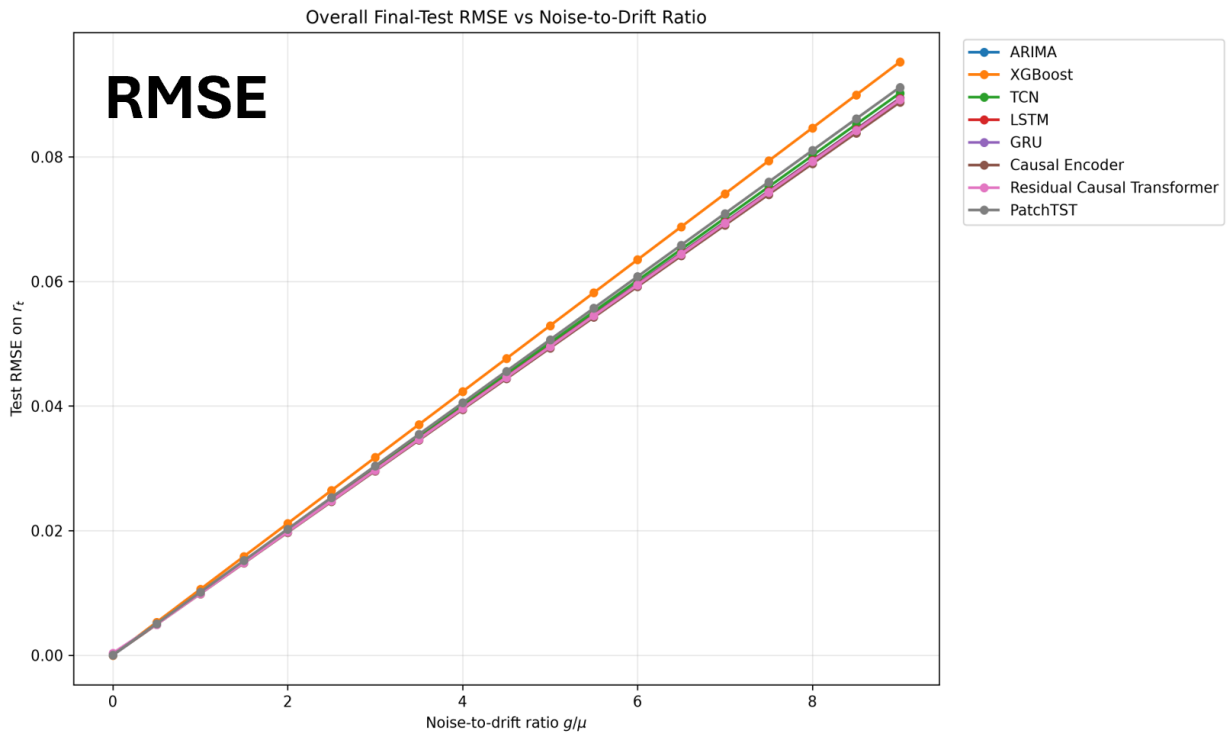
Fig. 24 – Some examples of generated synthetic time series

- $s = \frac{g}{\mu}$

Fig. 26 – Performance of Trained machines on stochastic processes versus increasing g



- ARIMA
- XGBoost
- TCN
- LSTM
- GRU
- Causal Encoder Transformer
- Residual Encoder Transformer
- PatchTST



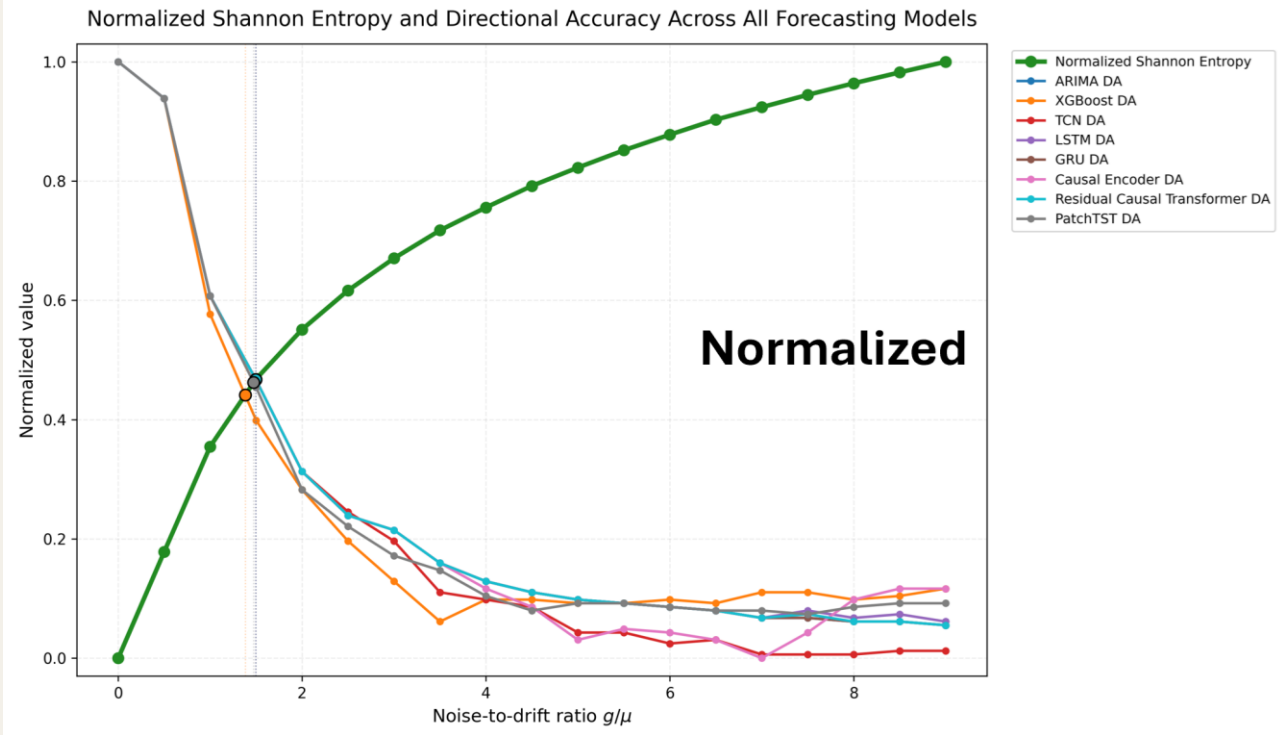


Fig. 27 – RMSE and Normalized Shannon Entropy and Directional Accuracy vs Noise-to-Drift Ratio

- Second Experiment

a = [+1, +1, +1, +1, +1, +1, +1, +1],
b = [+1, +1,−1, +1, +1, +1,−1,−1],
c = [+1,−1, +1,−1, +1,−1, +1, +1],
d = [+1,−1,−1,−1, +1,−1,−1,−1],
e = [−1,−1,−1,−1,−1,−1,−1,−1],
f = [−1,−1, +1,−1,−1,−1, +1, +1],
g = [−1, +1,−1, +1,−1, +1,−1,−1],
h = [−1, +1, +1, +1,−1, +1, +1, +1].

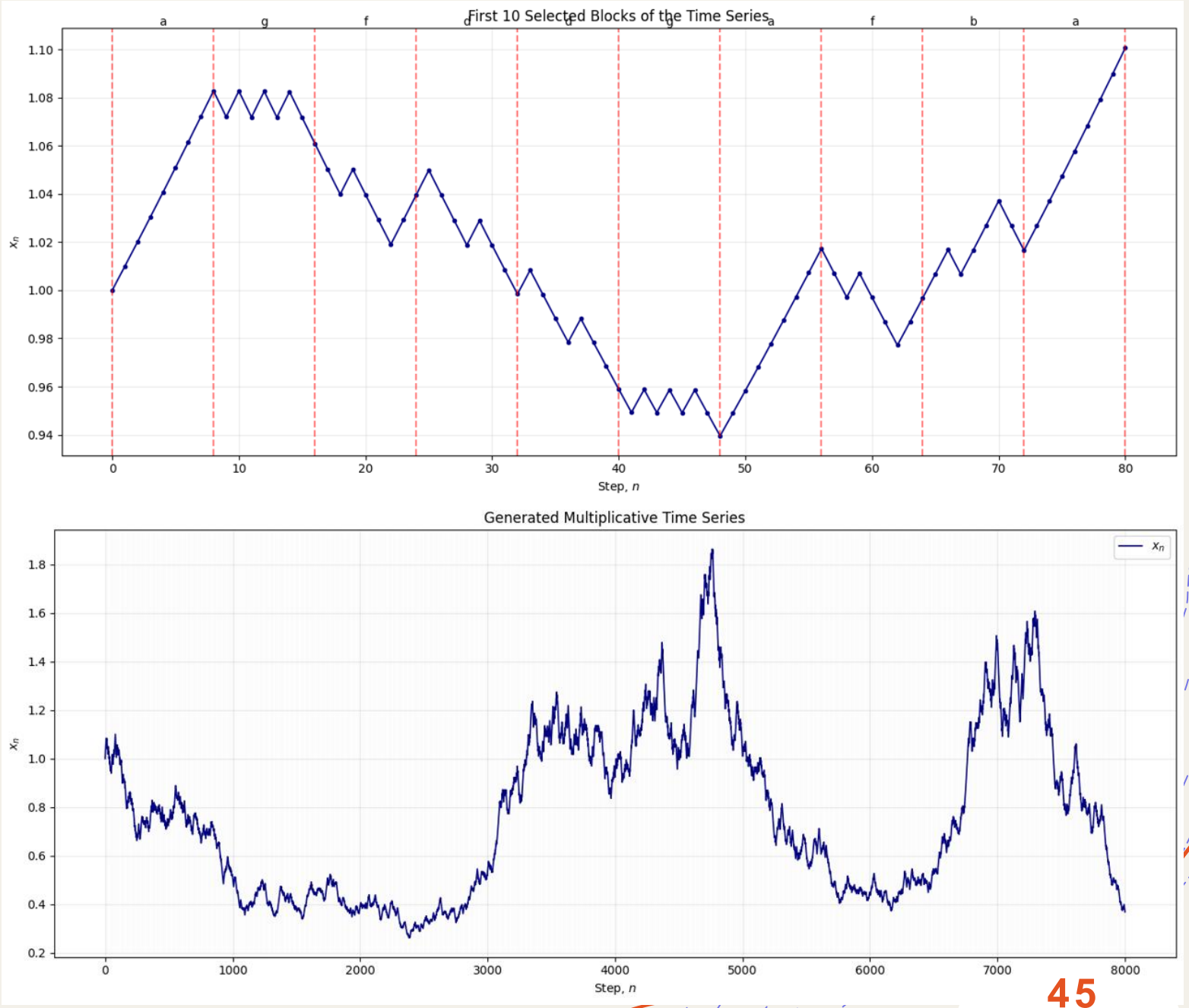


Fig. 28 – First 10 block of the time series and the whole generate using 1000 blocks

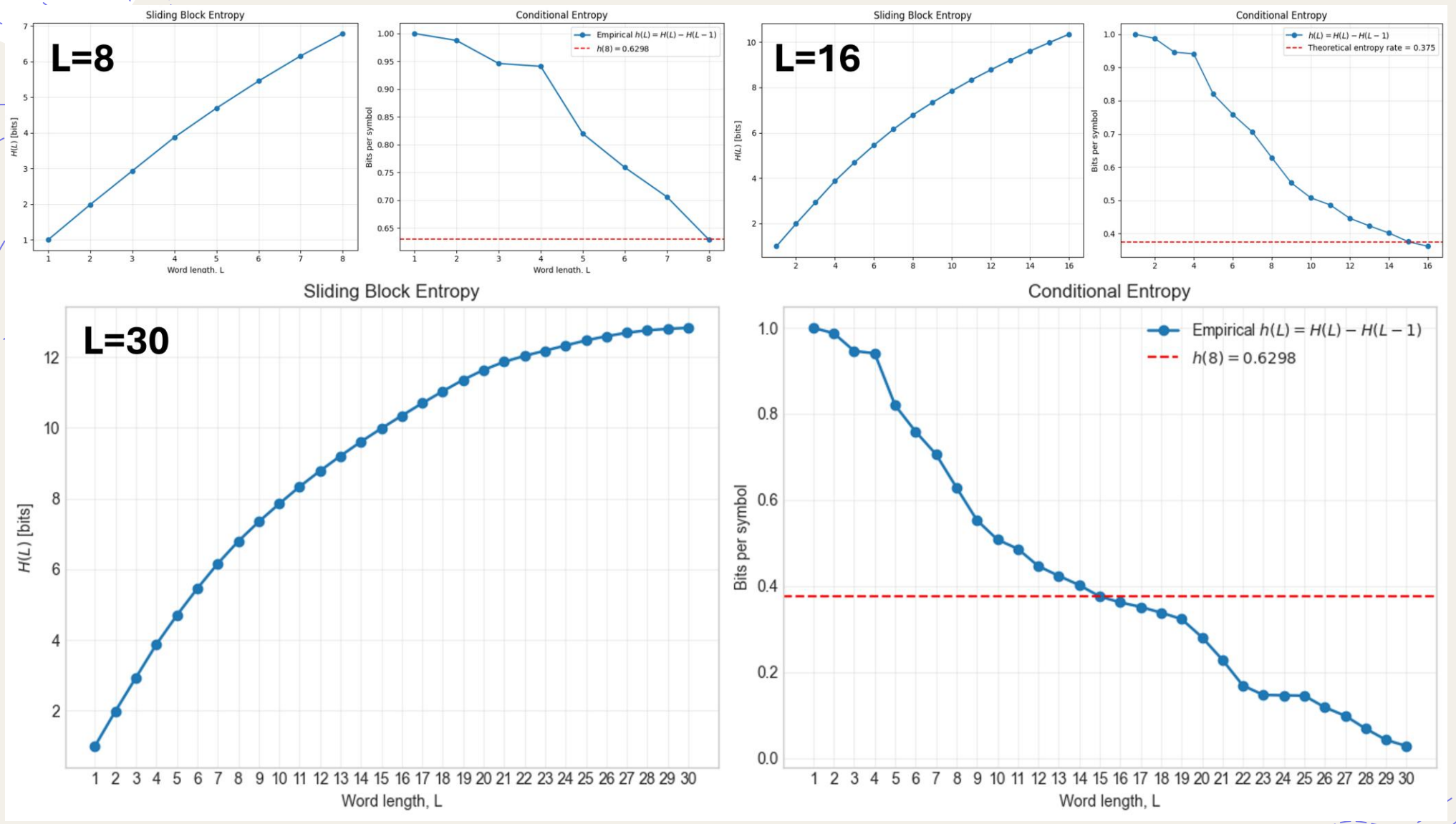


Fig. 29 – Block Entropy analysis over Word length, L

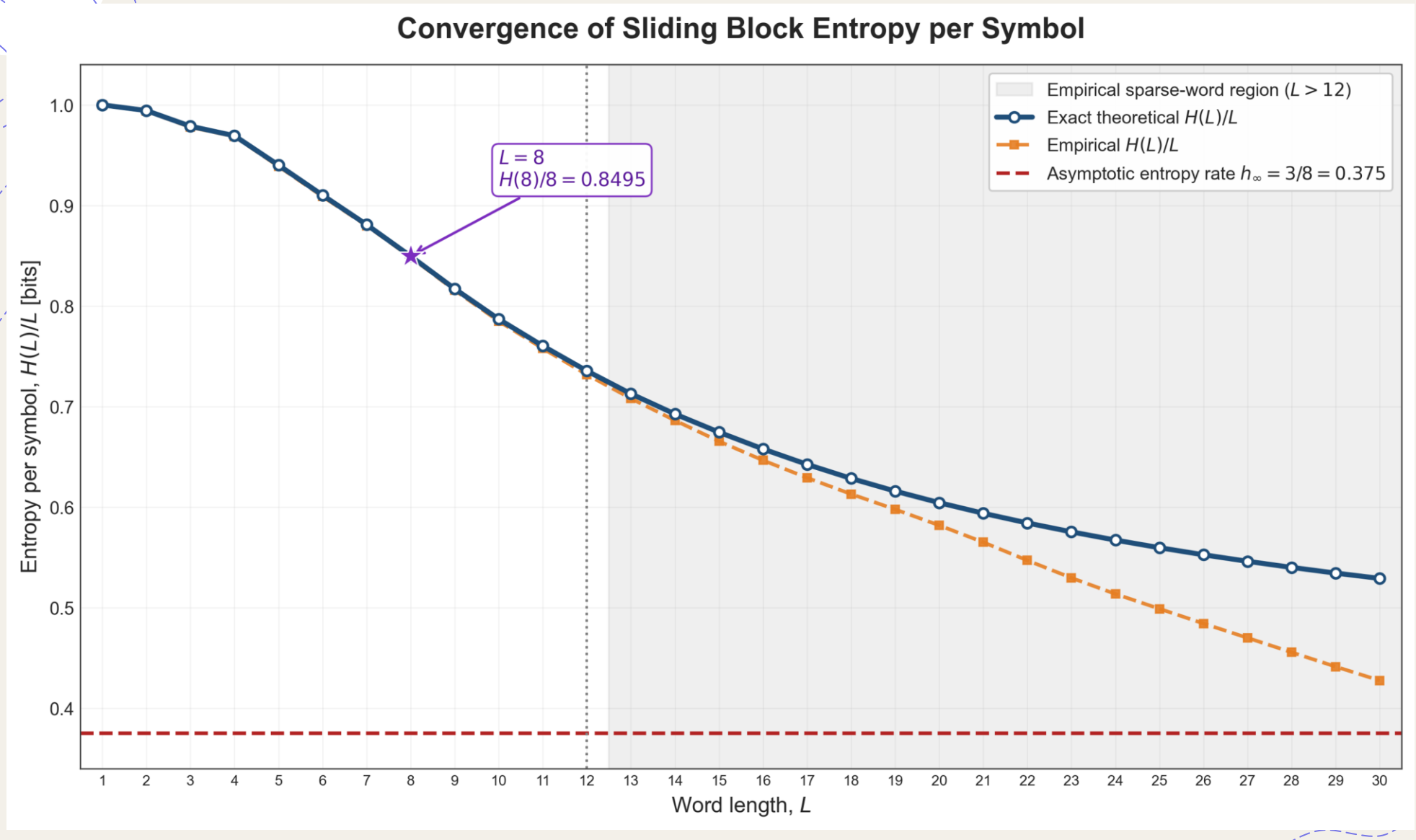


Fig. 30 – Convergence of Sliding Block Entropy per Symbol

- Experiment 2 – Regression:

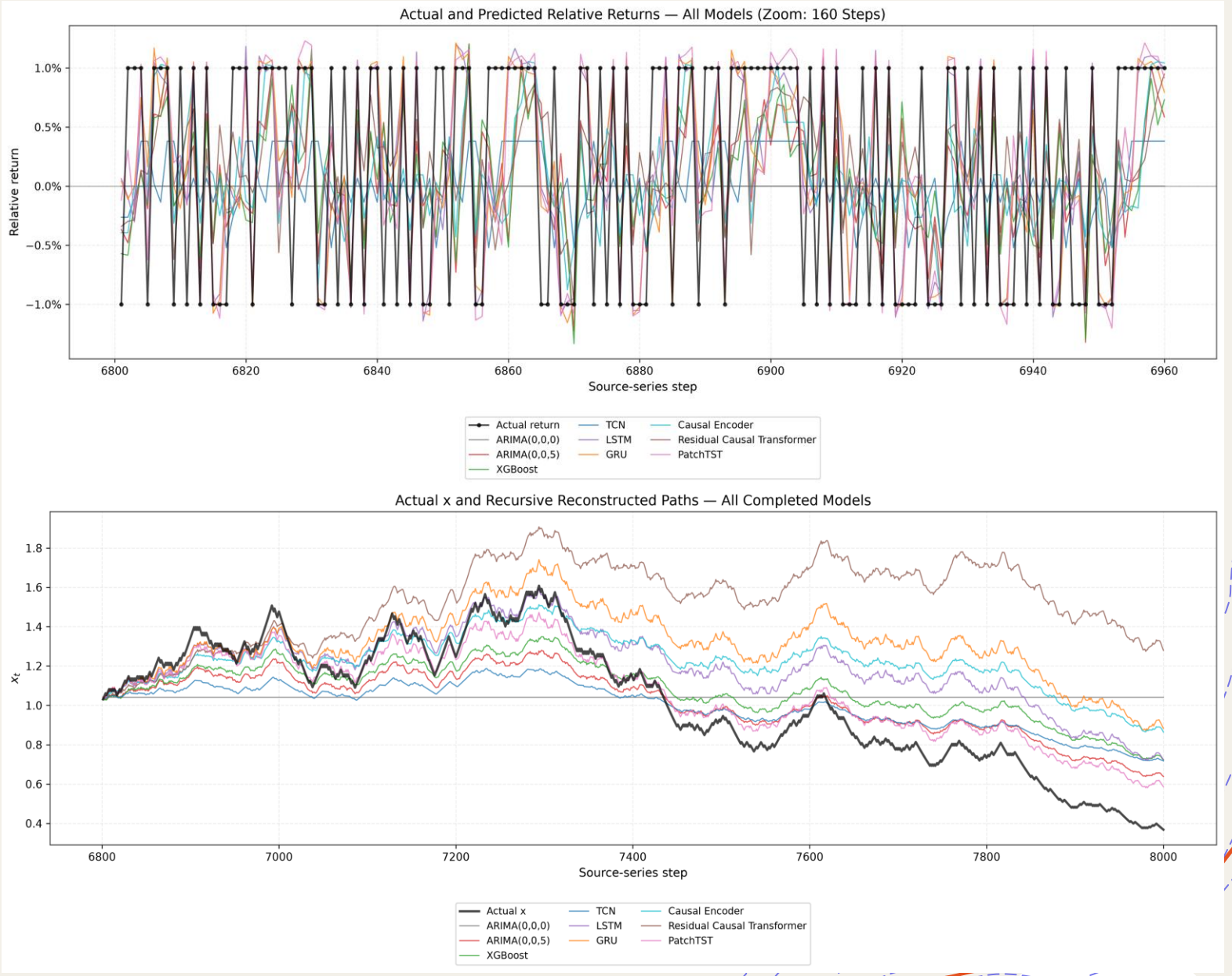


Fig. 31 – Return prediction and reconstructed paths for a Regression task

- Experiment 2 – Classification:

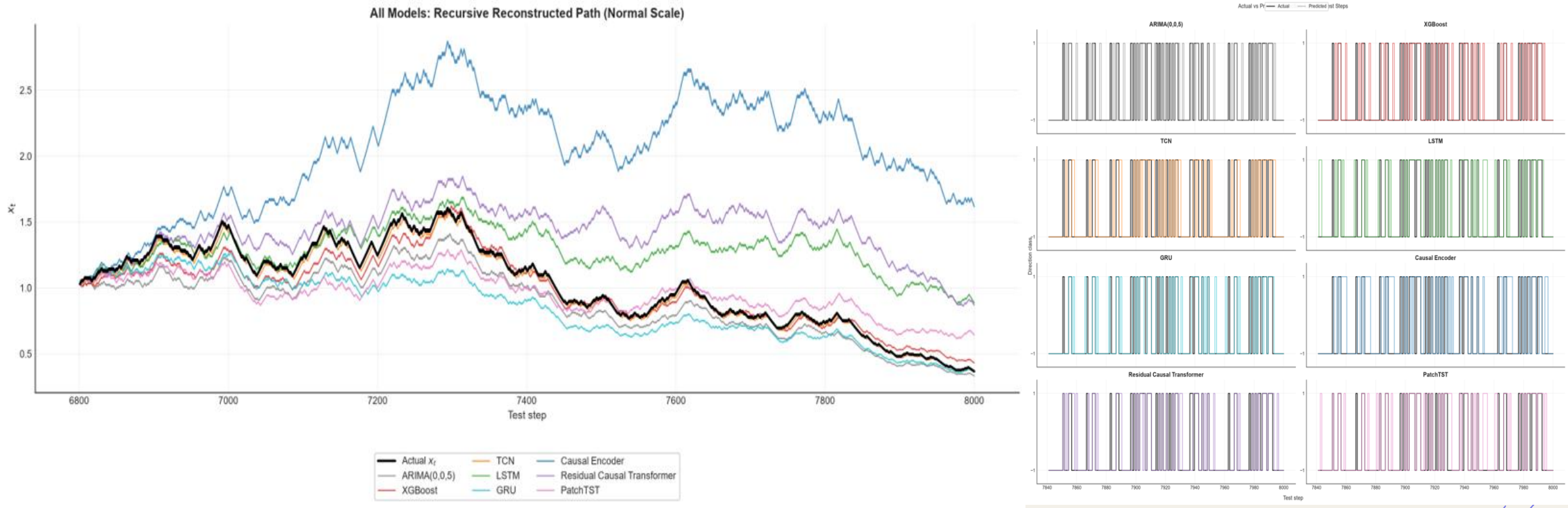


Fig. 31 – Return prediction and reconstructed paths for a Classification task

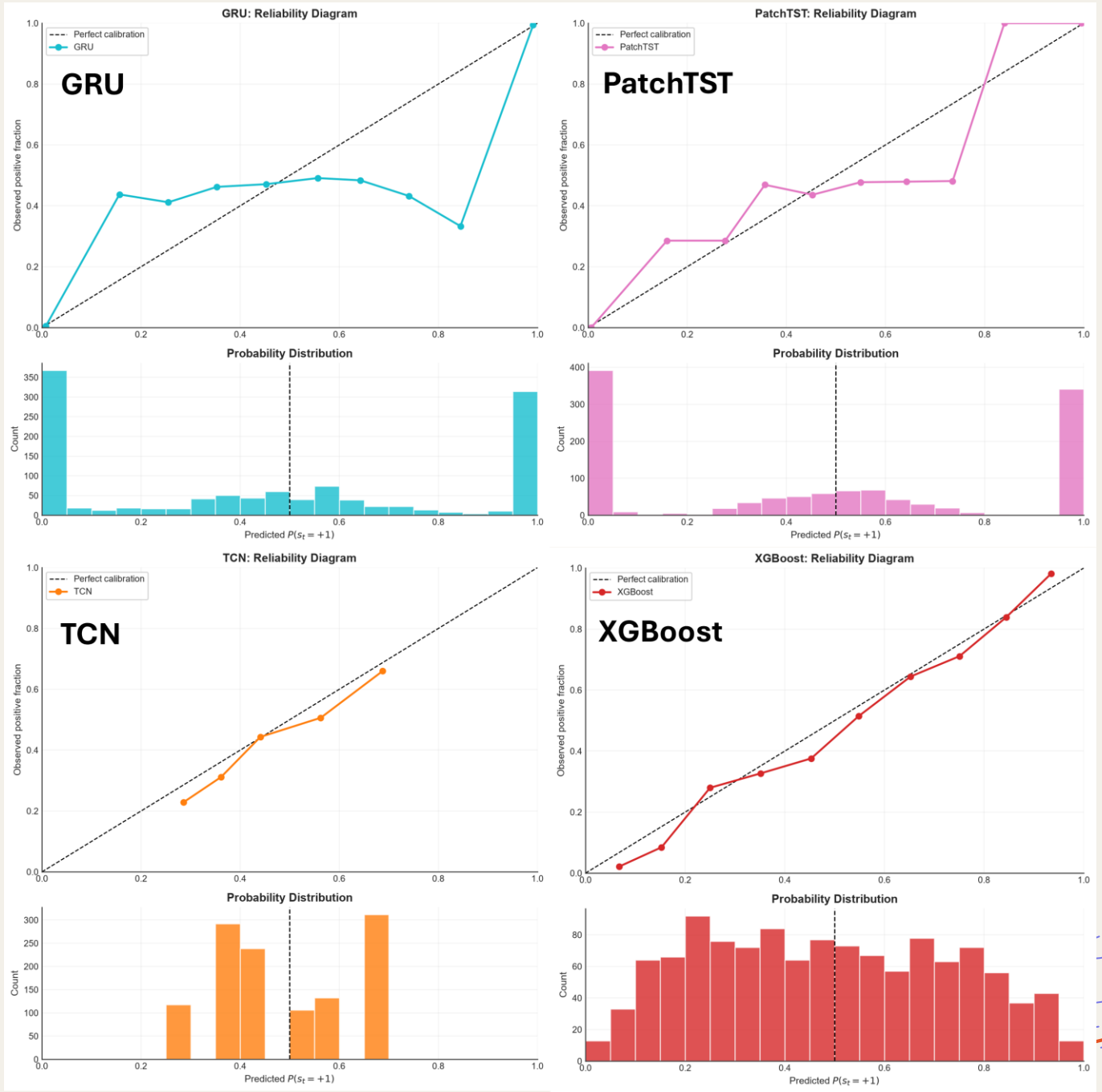


Fig. 32 – Reliability diagram of classification task

Table 2: Comparison of the performance of models on different task with the same data

Model	Regression DA	Classification DA	Change
ARIMA(0,0,5)	71.6%	71.6%	0.0 pp
XGBoost	72.3%	72.2%	−0.1 pp
TCN	62.7%	62.7%	0.0 pp
LSTM	78.5%	79.1%	+0.6 pp
GRU	78.6%	79.8%	+1.2 pp
Causal Encoder	71.8%	72.8%	+1.0 pp
Residual Causal Transformer	72.5%	80.2%	**+7.7 pp**
PatchTST	**81.0%**	**82.0%**	+1.0 pp

- Experiment 3:

$$RSI_t = 100 - \frac{100}{1 + \frac{n_+(t)}{n_-(t)}},$$

$$R_b = \frac{x_{t+8}^{(b)} - x_t}{x_t} = \prod_{j=1}^{8}\left(1 + \mu s_{b,j}\right) - 1.$$

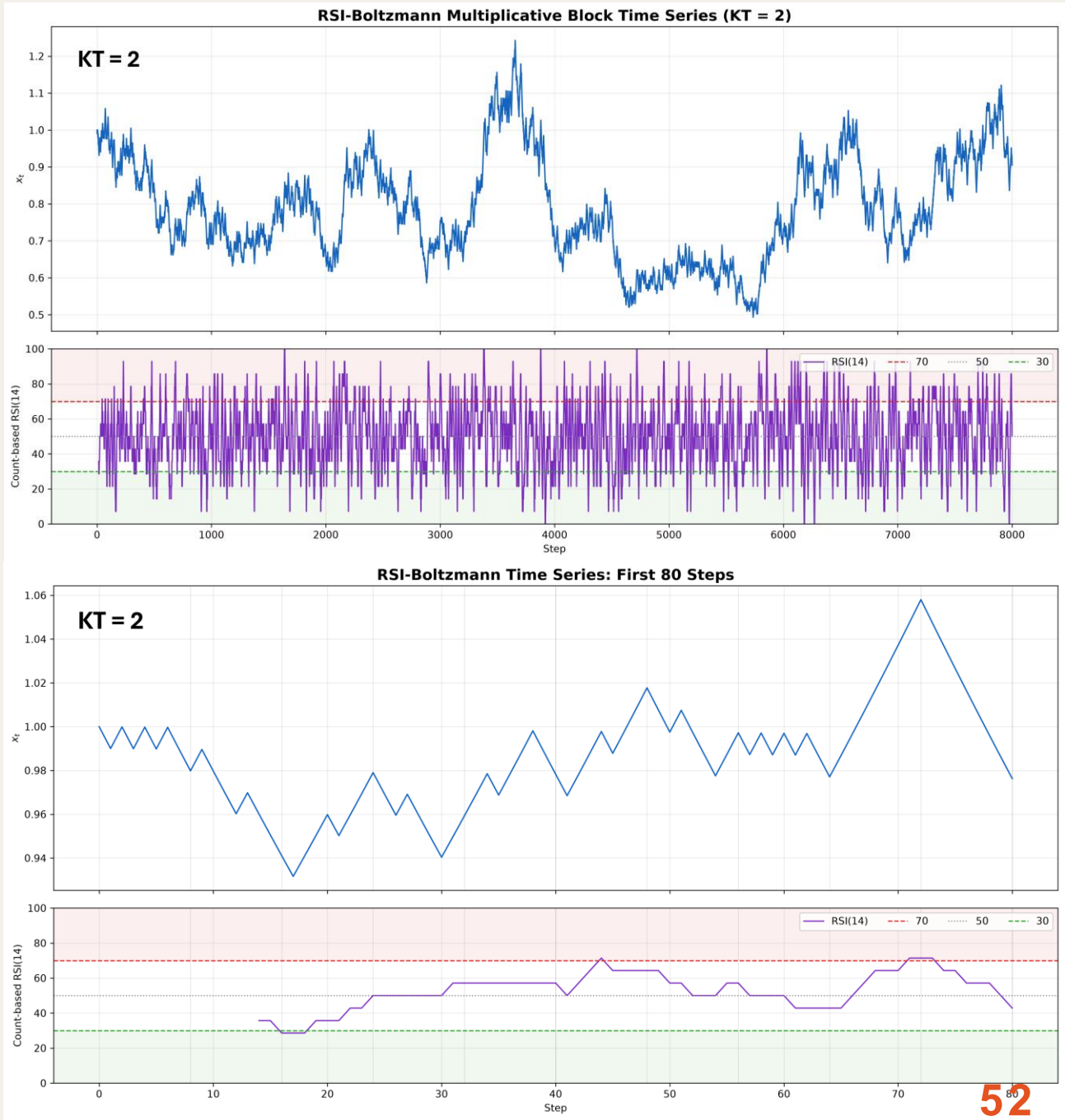


Fig. 33 – RSI-Boltzmann Time series

$$E_b(t) = (RSI_t - 50)R_b,$$

$$w_b(t) = \exp\left[-\frac{E_b(t)}{KT}\right].$$

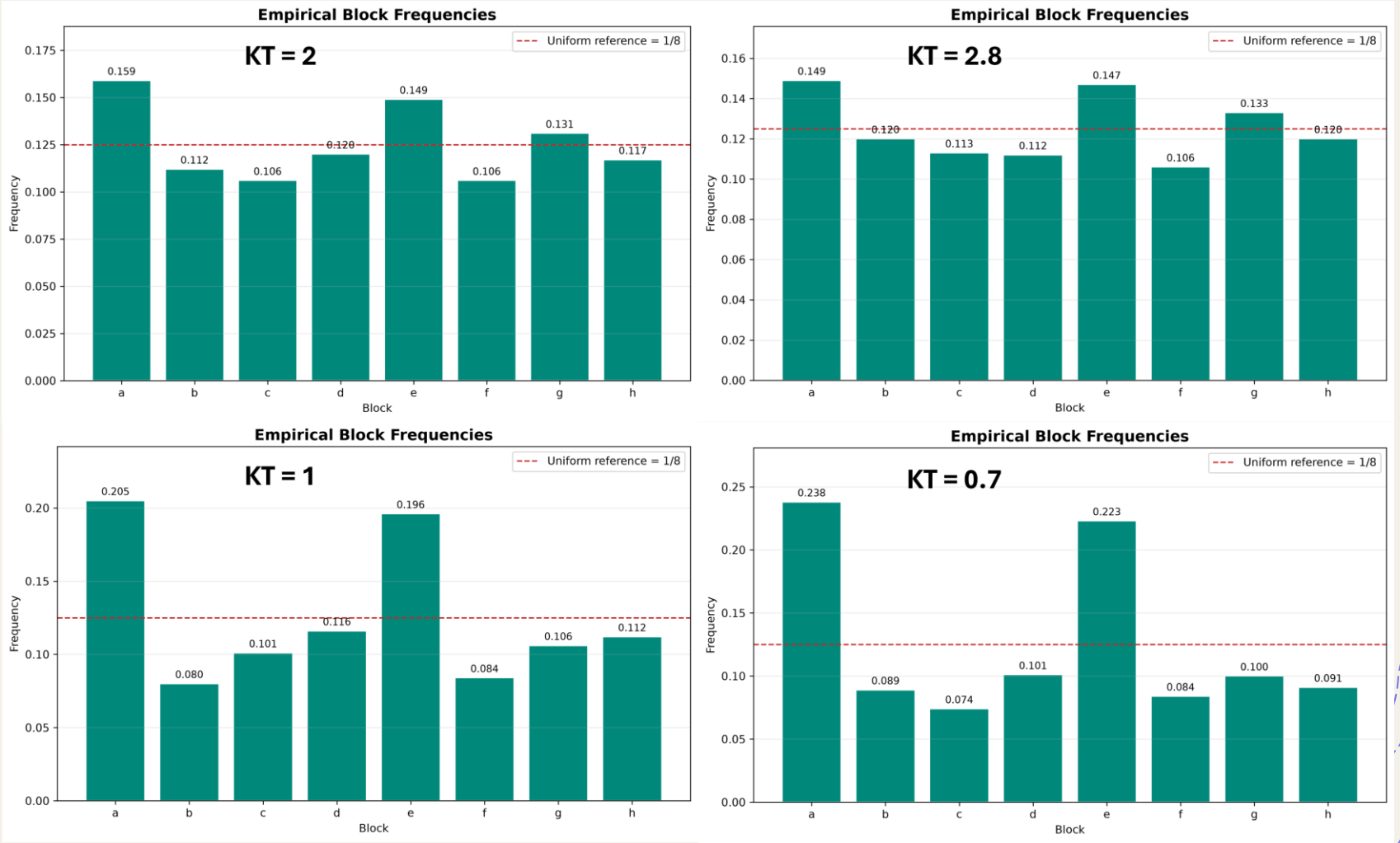


Fig. 34 – Empirical Block Frequencies

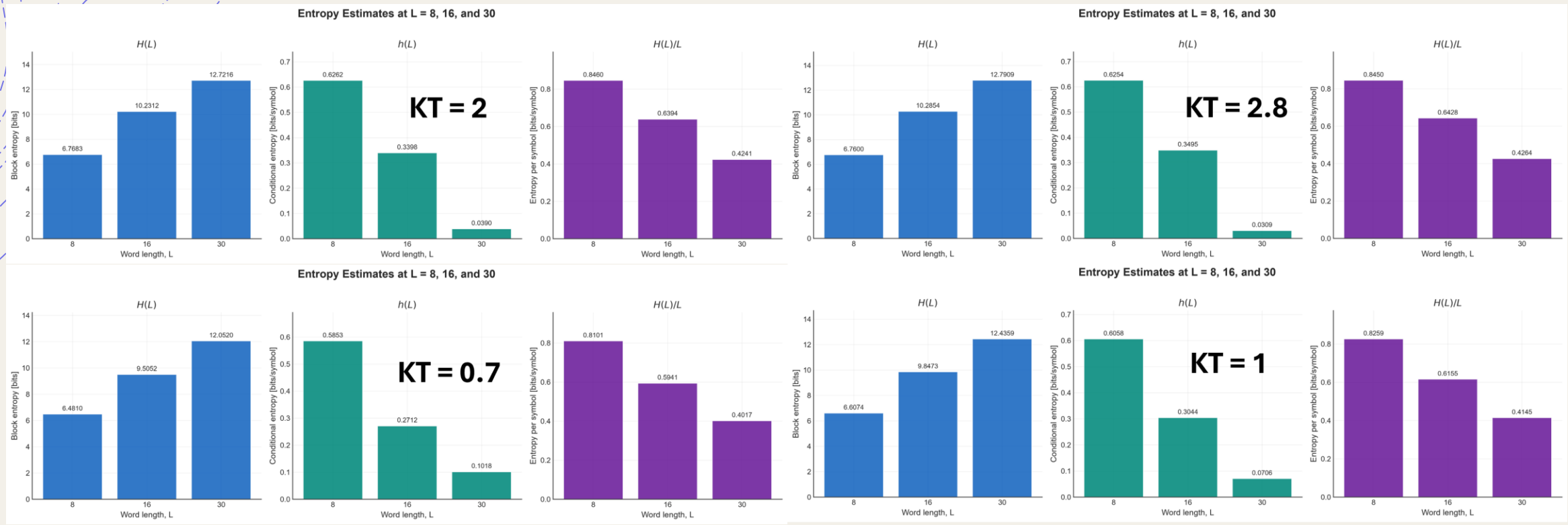


Fig. 31 – Entropy Estimates for different temperature parameters

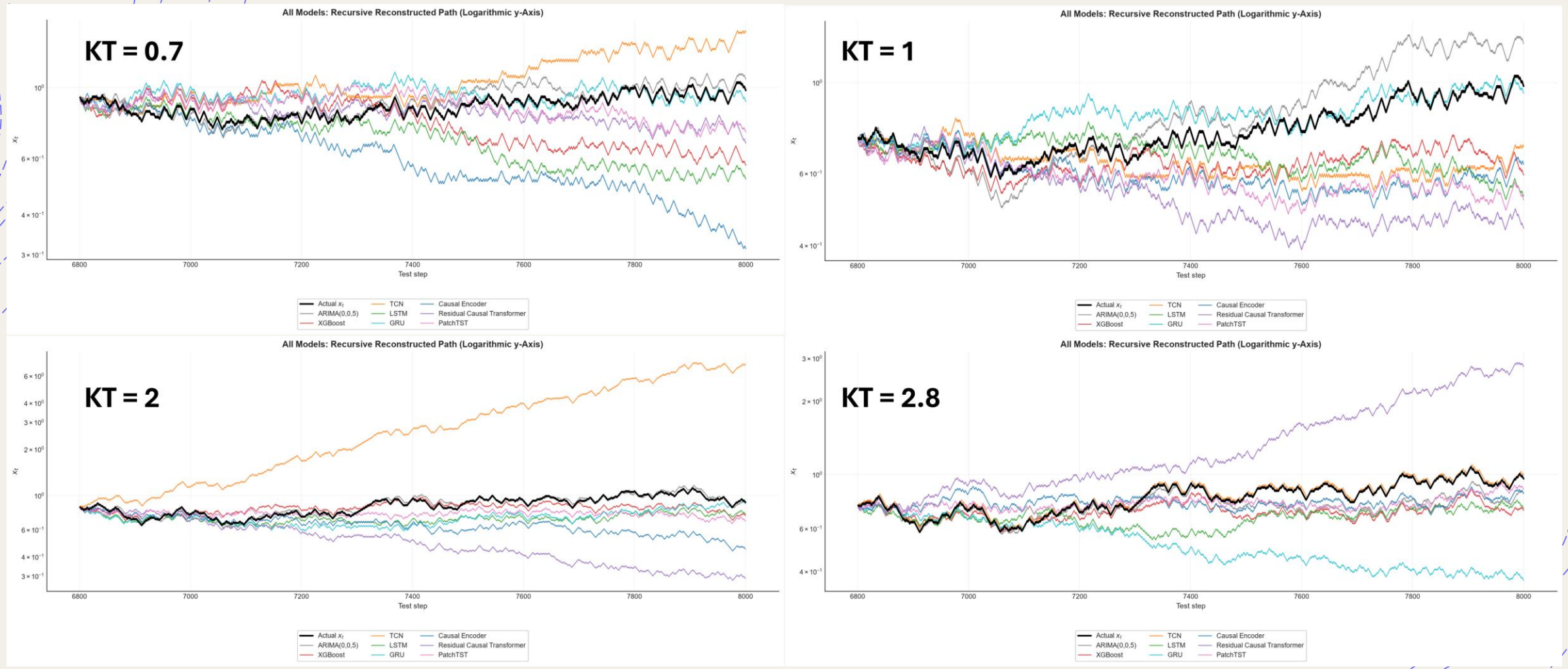


Fig. 32 – Reconstructed paths for RSI-Boltzmann Time series data

Table 3: Comparison of the performance of models on all experiments

Model	Regression-derived	Uniform direct	KT=2.8	KT=2	KT=1	KT=0.7
ARIMA(0,0,5)	71.4	71.4	71.9	71.4	71.0	71.7
XGBoost	72.1	72.1	70.3	72.1	71.8	74.1
TCN	62.4	62.4	64.3	62.4	62.1	61.8
LSTM	78.6	79.3	**81.2**	79.3	85.0	85.7
GRU	78.9	79.7	81.1	79.7	85.0	**86.3**
Causal Encoder	71.4	73.2	70.6	73.2	74.6	77.5
Residual Causal Transformer	73.2	80.4	75.6	80.4	81.4	78.6
PatchTST	**81.0**	**82.1**	80.8	**82.1**	**86.9**	85.9

- **Hints for Further Studies**

- Ising Model & Spin Glasses
- Multifractality on cross-correlations
- Multivariate models
- Hybrid Models
- Financial NLP & LLMs in Finance


The End